Building and evaluation of a PBPK model for Ketoconazole in adults

Version	**v1.0-OSP12.2**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Ketoconazole-Model/releases/tag/v1.0-OSP12.2
OSP Version	12.2
Qualification Framework Version	3.3

This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

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1 Introduction

Ketoconazole is an imidazole antifungal and a strong inhibitor of CYP3A4 and P-gp. Although oral ketoconazole use is restricted because of safety concerns, ketoconazole remains a clinically important reference perpetrator for CYP3A4-mediated and transporter-mediated drug-drug interaction assessments.

The model describes ketoconazole together with the circulating metabolites N-deacetylketoconazole and N-deacetyl-N-hydroxyketoconazole. The parent-metabolite structure was developed to capture ketoconazole pharmacokinetics after oral dosing and to quantify the contribution of parent and metabolites to CYP3A4 and P-gp inhibition Marok 2023. The model is used here to evaluate the oral ketoconazole pharmacokinetic data included in the model publication and to document the mechanistic assumptions relevant for DFI and DDI applications.

2 Methods

2.1 Modeling strategy

The general strategy for model development was based on a middle-out PBPK workflow. Drug-dependent parameters were taken from experimental data or estimated using established in silico methods where possible. Parameters that could not be fixed from independent evidence were optimized against clinical concentration-time profiles and then evaluated using independent oral dosing scenarios.

The model was implemented as a parent-metabolite model for ketoconazole, N-deacetylketoconazole and N-deacetyl-N-hydroxyketoconazole. Ketoconazole plasma profiles from oral solution, capsule-as-solution and tablet studies were used to inform absorption, dissolution and disposition parameters, while the Weiss et al. clinical data provided the only plasma concentration-time profiles for N-deacetylketoconazole Weiss 2022.

Ketoconazole absorption was evaluated across fasted and fed dosing conditions, including solution, capsule-as-solution and tablet formulations. The model includes CYP3A4-mediated metabolism, arylacetamide deacetylase-mediated N-deacetylketoconazole formation, UGT1A4-mediated metabolism, P-gp transport and reversible inhibition of CYP3A4 and P-gp by ketoconazole and metabolites. The resulting model structure supports both concentration-time profile evaluation and DFI or DDI perpetrator simulations Marok 2023.

2.2 Data used

Clinical plasma concentration-time profiles were taken from the published studies compiled for the model publication Marok 2023. The dataset covers oral single-dose and multiple-dose ketoconazole administration in adults, fasted and fed states, and doses from 100 mg to 1200 mg.

Publication	Data used
Heel 1982	Oral solution and tablet ketoconazole studies in adults under fasted conditions.
Daneshmend 1981, Daneshmend 1983, Daneshmend 1984	Oral ketoconazole studies in adults covering fasted, fed, single-dose and multiple-dose administration.
Huang 1986	Oral solution and tablet ketoconazole data in healthy adults.
Polk 1999, Boyce 2012, Tiseo 1998, Craven 1983, Männistö 1982, Greenblatt 1998	Oral ketoconazole clinical data used to evaluate formulation, food-effect and multiple-dose performance.
Weiss 2022	Oral ketoconazole study in adults with plasma profiles for ketoconazole and N-deacetylketoconazole.

Physicochemical, transporter and metabolic information was taken from experimental literature and in silico calculations. Ketoconazole and metabolite pK_a, lipophilicity and solubility parameters were informed by Chemicalize and formulation-relevant dissolution data, while in vitro metabolism and inhibition data were taken from studies on AADAC, CYP3A4, P-gp and UGT pathways Fukami 2016, Fitch 2009, Schwab 2003, Bourcier 2010, Weiss 2022.

The FMO3 K_m value of 1.17 µmol/L implemented for N-deacetylketoconazole is correct. The corresponding value reported by Marok 2023 is erroneous.

2.3 Model parameters and assumptions

Absorption and formulations

The model includes oral solution, capsule-as-solution and tablet applications. Solution and capsule-as-solution scenarios were used to separate systemic disposition from dissolution-limited tablet absorption. Tablet absorption was described with formulation-specific dissolution and particle-size assumptions, with fed-state scenarios used to describe the reduced and delayed absorption observed for weak-base ketoconazole under altered gastrointestinal conditions Marok 2023, Table 1.

Distribution

Ketoconazole is a weak base with high plasma protein binding and pH-dependent solubility. Distribution parameters were based on physicochemical properties and PBPK tissue partitioning methods, with optimized lipophilicity where required to reproduce systemic exposure. The metabolites were represented as separate compounds with their own physicochemical properties and protein-binding assumptions Marok 2023, Table 1.

Metabolism, transport and inhibition

Ketoconazole metabolism includes formation of N-deacetylketoconazole by arylacetamide deacetylase and further metabolism through CYP3A4 and UGT pathways. Ketoconazole and metabolites were represented as CYP3A4 and P-gp inhibitors, which is required to describe the observed DDI perpetrator behavior more accurately than a parent-only inhibition model Marok 2023, Table 1, and Weiss 2022.

3 Results and Discussion

The concentration-time profiles cover oral solution, capsule-as-solution and tablet administration under fasted and fed conditions. The model reproduced the broad exposure range across doses and formulations, including the delayed and reduced absorption observed in fed tablet scenarios.

The ketoconazole goodness-of-fit diagnostics summarize the parent plasma data across the full clinical dataset. N-deacetylketoconazole diagnostics are shown separately because metabolite observations are only available from the Weiss et al. study. This split avoids mixing parent and metabolite observations in a single legend and keeps analyte-specific performance interpretable.

In the model publication, the parent-metabolite model described 53 plasma concentration-time profiles, with 49/53 profiles, 50/53 AUC_{last} values and 52/53 C_{max} values within twofold Marok 2023, Table 2. DFI performance was adequate across the evaluated food-effect ratios, with 7/7 AUC_{last} and C_{max} ratios within the reported acceptance limits Marok 2023, Table 3. For DDI simulations, including ketoconazole and both metabolites as CYP3A4 and P-gp inhibitors improved perpetrator performance compared with parent-only inhibition Marok 2023.

3.1 Ketoconazole final input parameters

The following tables are generated from the model and summarize the final compound input parameters for ketoconazole, N-deacetylketoconazole and N-deacetyl-N-hydroxyketoconazole. Optimized parameters should be interpreted together with the source references and model assumptions described in Section 2.3.

The particle-radius unit reported in Table S1.3 of the supplement is erroneous. The intended radii are 11.749, 111.06 and 205.46 µm for tablet dissolution bins 1, 2 and 3, respectively. The corresponding snapshot values of 0.011749, 0.11106 and 0.20546 mm are correct. The solubility value shown below represents a pH-dependent table function rather than a single constant.[1]

Compound: Ketoconazole

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility table	12872590000000 mg/l	Publication-In Vitro-Extrapolated from observed data	Ghazal 2015	True
Lipophilicity	2.5243939623 Log Units	Parameter Identification-Parameter Identification-Optimized	fit	True
Fraction unbound (plasma, reference value)	0.01	Publication-Other	Heel 1982	True
Specific intestinal permeability (transcellular)	1.5597039897E-05 cm/min	Parameter Identification-Parameter Identification-Optimized	Fit fasted	True
Specific intestinal permeability (transcellular)	9.9497068191E-06 cm/min	Parameter Identification-Parameter Identification-Optimized	Fit fed	False
Cl	2	Internet-Other-Chemicalize		
Is small molecule	Yes			
Molecular weight	531.43 g/mol	Internet-Other-Chemicalize		
Plasma protein binding partner	Albumin			
Treat precipitated drug as	Insoluble	Unknown		
Aqueous diffusion coefficient	3.7482784124E-07 dm^2/min	Parameter Identification-Parameter Identification-Optimized		
Density (drug)	1.4 g/cm^3	Internet-Other-ChemSpider		
Enable supersaturation	Yes	Unknown		

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	PK-Sim Standard

Processes

Metabolizing Enzyme: CYP3A4-Assumed

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	0.008461538 µmol/l	Other-Assumption-Assumed from Ki value from Weiss 2022
kcat	0.1020204084 1/min	Parameter Identification-Parameter Identification-Optimized

Transport Protein: ABCB1-Assumed

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Transporter concentration	1 µmol/l	
Vmax	0 µmol/l/min	
Km	0.035 µmol/l	Other-Assumption-Assumed from Ki value from Weiss 2022
kcat	0.326999715 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-GFR

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Metabolizing Enzyme: UGT1A4-Bourcier 2010

Molecule: UGT1A4

Parameters

Name	Value	Value Origin
In vitro Vmax/recombinant enzyme	0 nmol/min/pmol rec. enzyme	
Km	7 µmol/l	Publication-In Vitro-recombinant enzymes
kcat	0.3062809592 1/min	Parameter Identification-Parameter Identification-Optimized

Inhibition: CYP3A4-Weiss 2022

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Ki	0.008461538 µmol/l	Publication-In Vitro-Calculated from IC50 and corrected for fumic

Metabolizing Enzyme: AADAC-Fukami 2016

Molecule: AADAC

Metabolite: N-deacetylketoconazole

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	28.3 pmol/min/mg mic. protein	
Km	1.8797 µmol/l	Publication-In Vitro-Corrected with $f_{u,mic}$ (17.5*0.107)
kcat	0.8694012741 1/min	Parameter Identification-Parameter Identification-Optimized

Inhibition: ABCB1-Weiss 2022

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Ki	0.035238095 µmol/l	Publication-In Vitro-Calculated from IC50 value

Compound: N-deacetylketoconazole

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1.24 mg/ml	Internet-Other-Chemicalize	Chemicalize	True
Reference pH	6.5	Internet-Other-Chemicalize	Chemicalize	True
Lipophilicity	3.7501327278 Log Units	Parameter Identification-Parameter Identification-Optimized	Fit	True
Fraction unbound (plasma, reference value)	1 %	Other-Assumption-assumed from value for Ketoconazole	Assumed	True
Cl	2	Internet-Other-Chemicalize		
Is small molecule	Yes			
Molecular weight	489.4 g/mol	Internet-Other-Chemicalize		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	Charge dependent Schmitt

Processes

Inhibition: CYP3A4-Weiss 2022

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Ki	0.022 µmol/l	Publication-In Vitro-Calculated from IC50 and corrected for fumic

Systemic Process: Glomerular Filtration-GFR

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Metabolizing Enzyme: FMO3-Rodriguez 1997

Molecule: FMO3

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	1.167401215 µmol/l	Publication-In Vitro-recombinant enzymes; model value is correct, publication value is erroneous
kcat	378.6506200467 1/min	Parameter Identification-Parameter Identification-Optimized

Inhibition: ABCB1-Weiss 2022

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Ki	0.119047619 µmol/l	Publication-In Vitro-Calculated from IC50

Compound: N-deacetyl-N-hydroxyketoconazole

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.0044 mg/ml	Internet-Other-Chemicalize	Chemicalize	True
Reference pH	6.5	Internet-Other-Chemicalize	Chemicalize	True
Lipophilicity	4.2 Log Units	Internet-Other-Chemicalize	Chemicalize	True
Fraction unbound (plasma, reference value)	1 %	Publication-Assumption-assumed from value for Ketoconazole	Assumed	True
Permeability	0 cm/min		Assumed	True
Cl	2	Internet-Other-Chemicalize		
Is small molecule	Yes			
Molecular weight	505.4 g/mol	Internet-Other-Chemicalize		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Berezhkovskiy
Cellular permeabilities	Charge dependent Schmitt

Processes

Inhibition: CYP3A4-Weiss 2022

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Ki	0.022 µmol/l	Other-Assumption-assumed from value for N-deacetylketoconazole

Systemic Process: Glomerular Filtration-GFR

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Inhibition: ABCB1-Weiss 2022

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Ki	0.119047619 µmol/l	Other-Assumption-assumed from value for N-deacetylketoconazole

Metabolizing Enzyme: FMO3-Clearance

Molecule: FMO3

Parameters

Name	Value	Value Origin
In vitro CL/recombinant enzyme	0 µl/min/pmol rec. enzyme	
CLspec/[Enzyme]	0.0893329451 l/µmol/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

The goodness-of-fit diagnostics combine all modeled compounds in one set of plots. Colors and symbols identify compounds consistently with the concentration-time profiles. Administration route, formulation, and model-building or verification status are not used to split the diagnostics.

Table 3-1: GMFE for Observed versus simulated concentration-time data for all modeled compounds

Group	GMFE
Ketoconazole	1.71
N-deacetylketoconazole	1.93
All	1.72

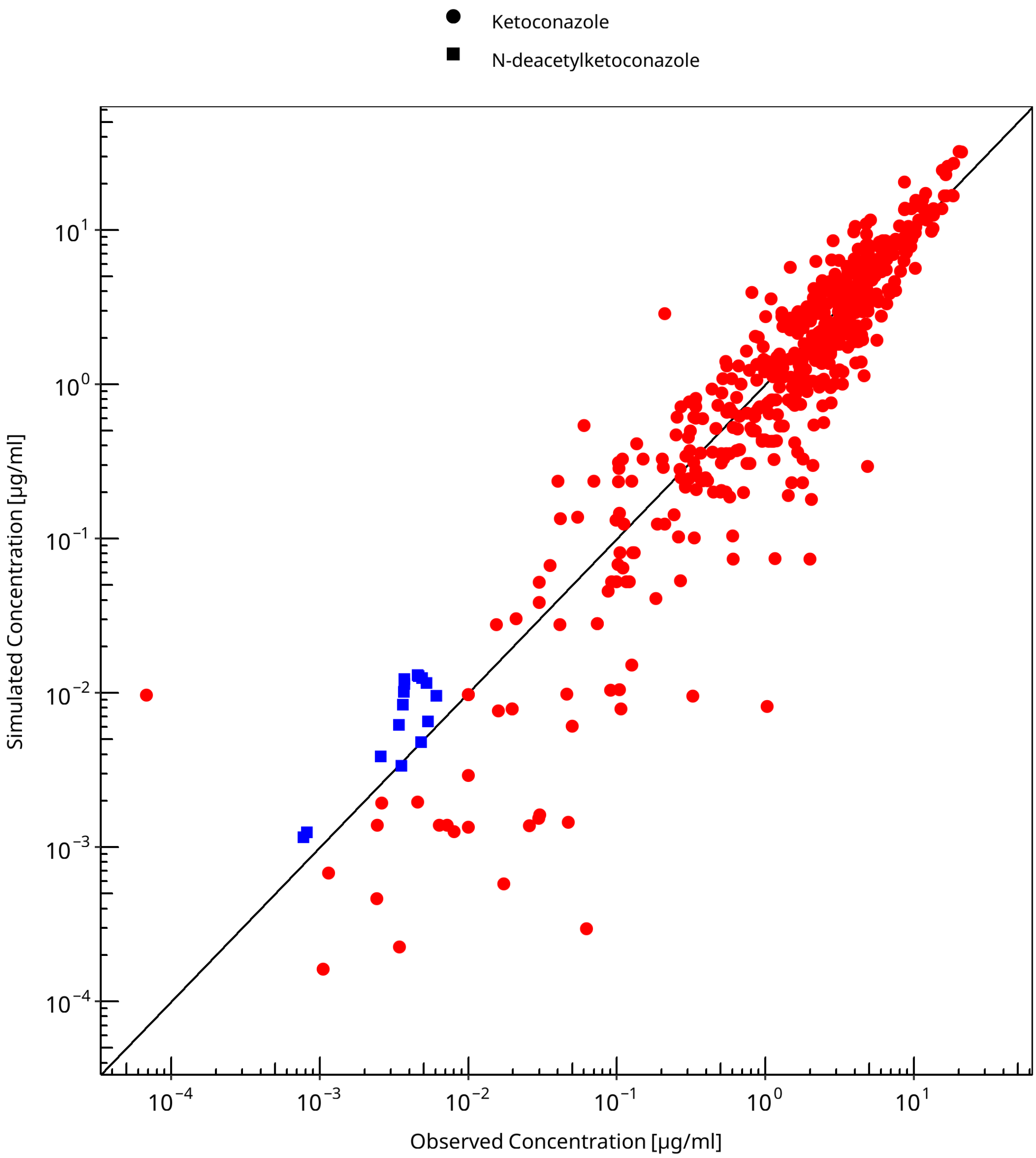


Figure 3-1: Observed versus simulated concentration-time data for all modeled compounds

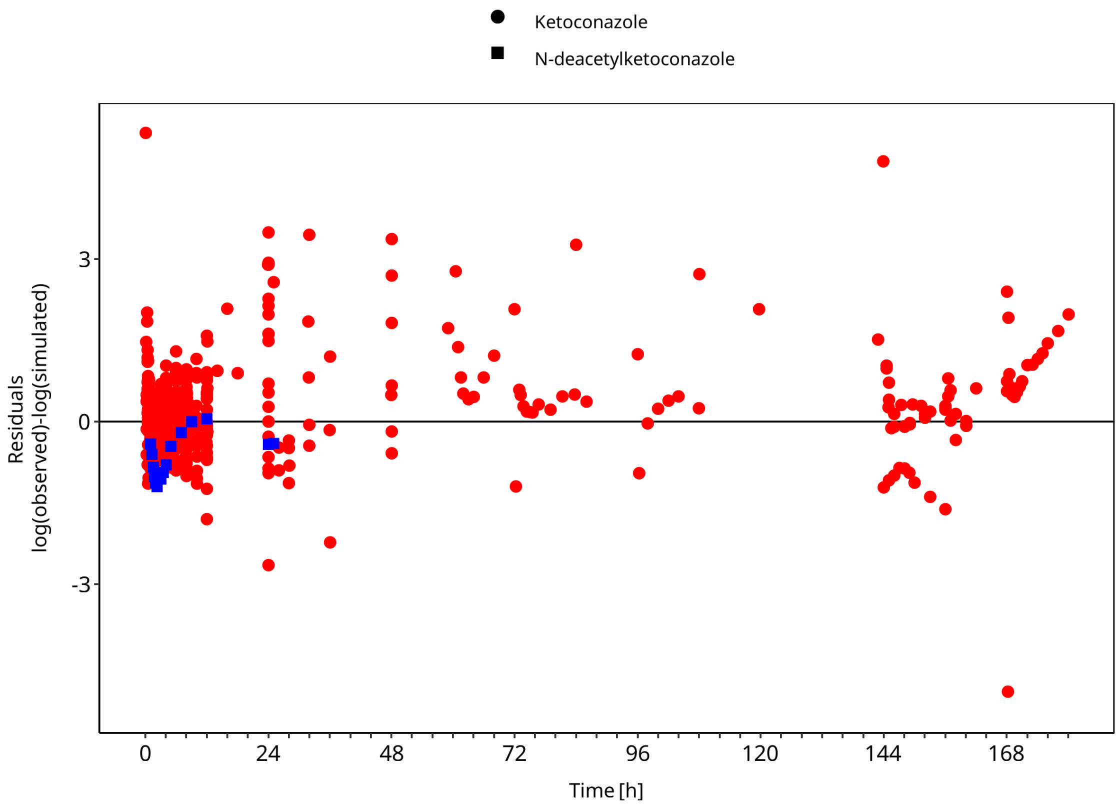


Figure 3-2: Observed versus simulated concentration-time data for all modeled compounds

3.3 Concentration-time profiles

Concentration-time profiles are grouped by formulation and dosing context. Solution and capsule-as-solution scenarios are shown separately from tablet studies because they inform different absorption assumptions.

3.3.1 Solution and capsule-as-solution studies

This section contains oral solution and capsule-as-solution studies used to evaluate absorption without tablet dissolution as the dominant limitation.

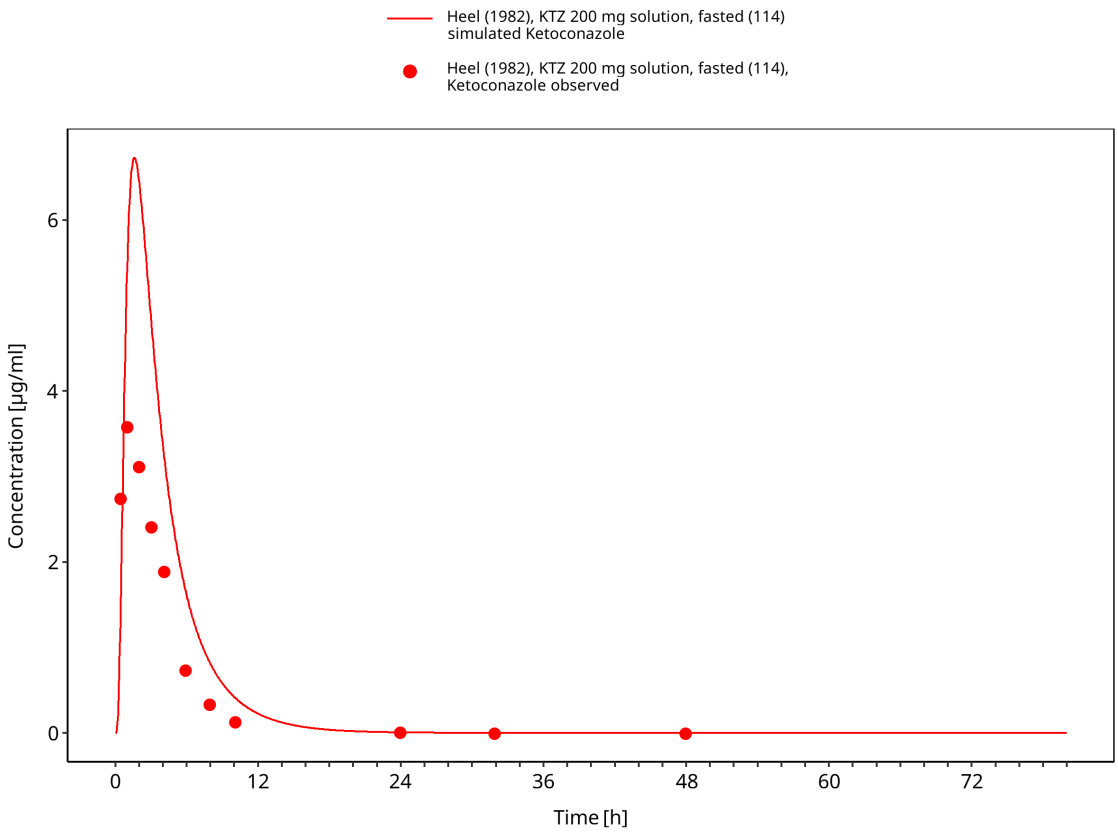


Figure 3-3: Time Profile Analysis

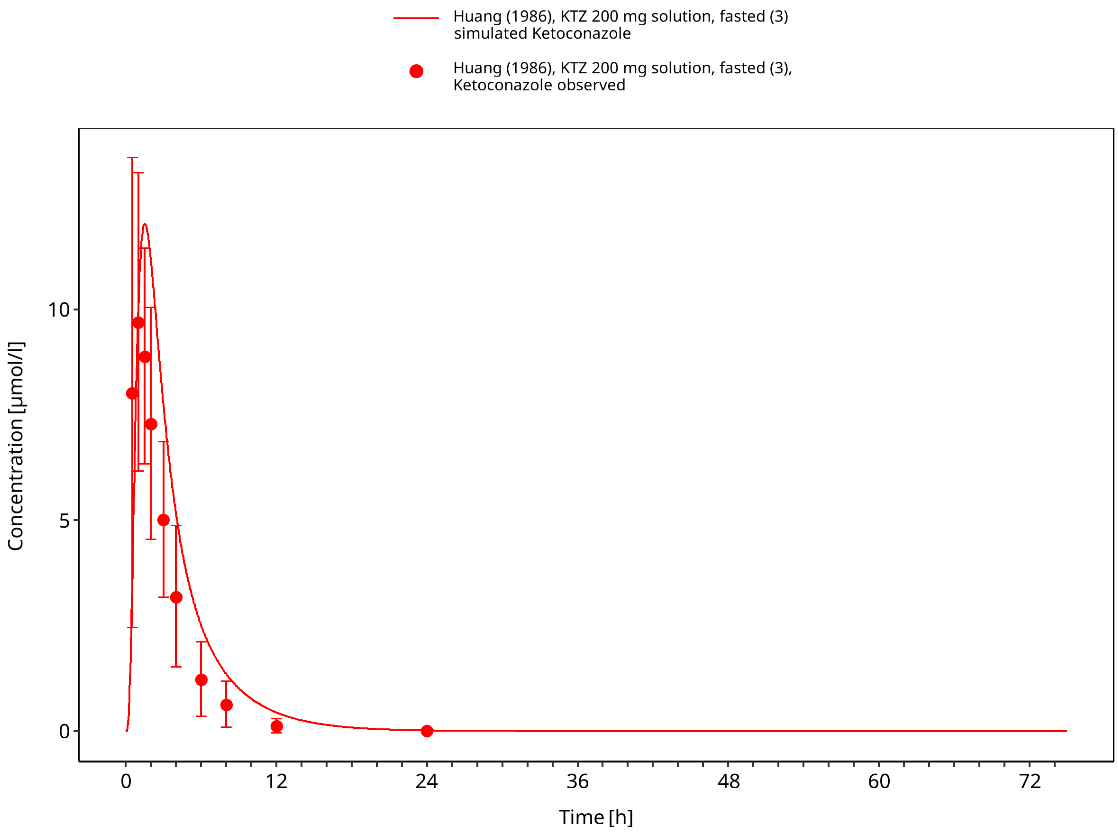


Figure 3-4: Time Profile Analysis

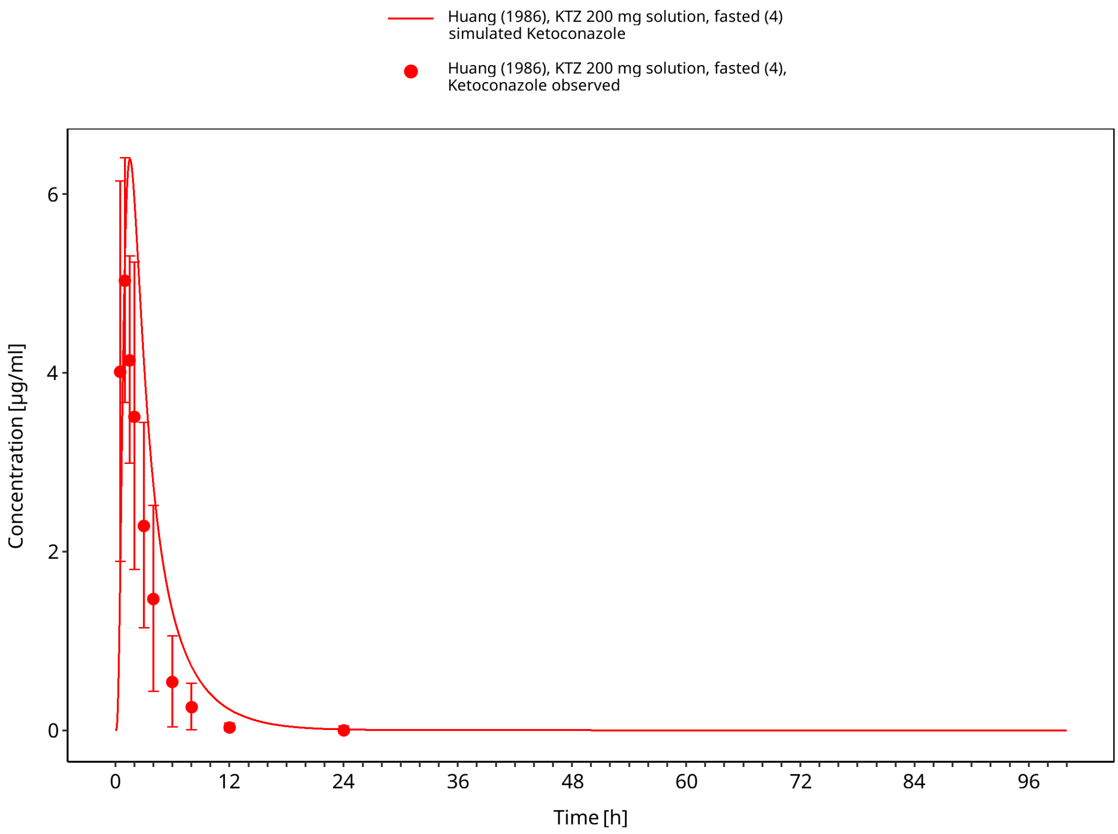


Figure 3-5: Time Profile Analysis

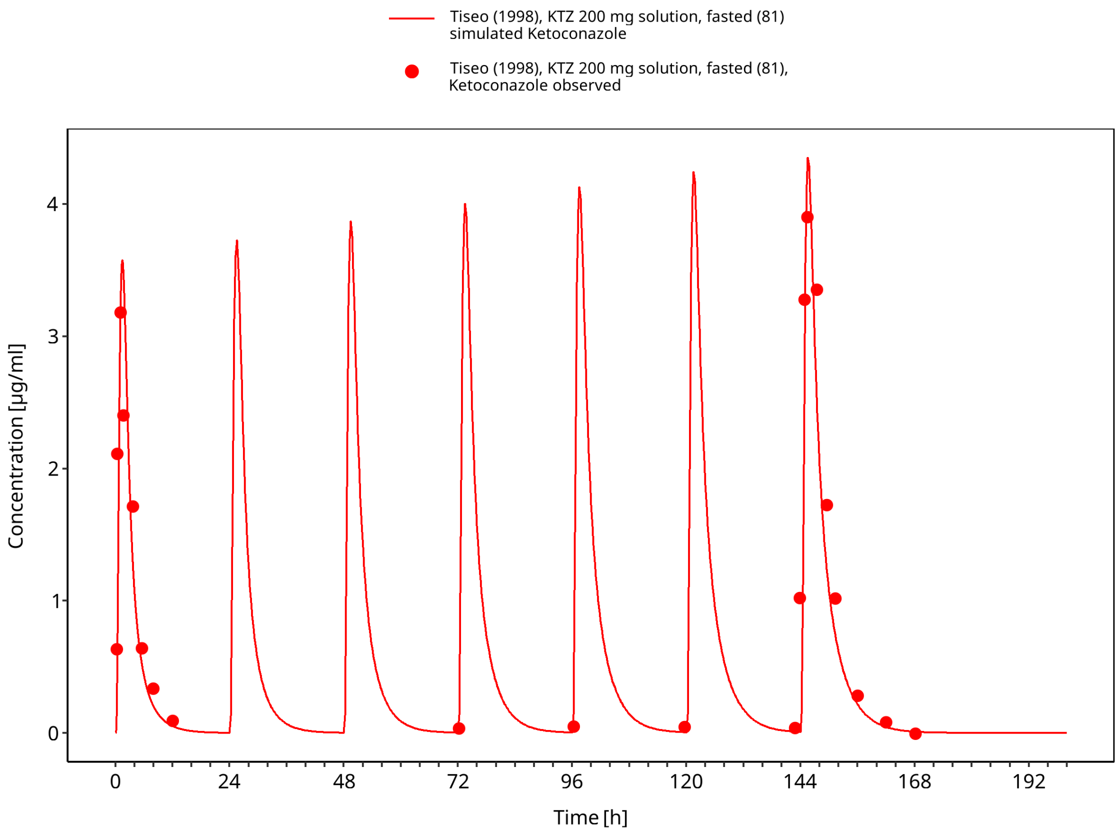


Figure 3-6: Time Profile Analysis

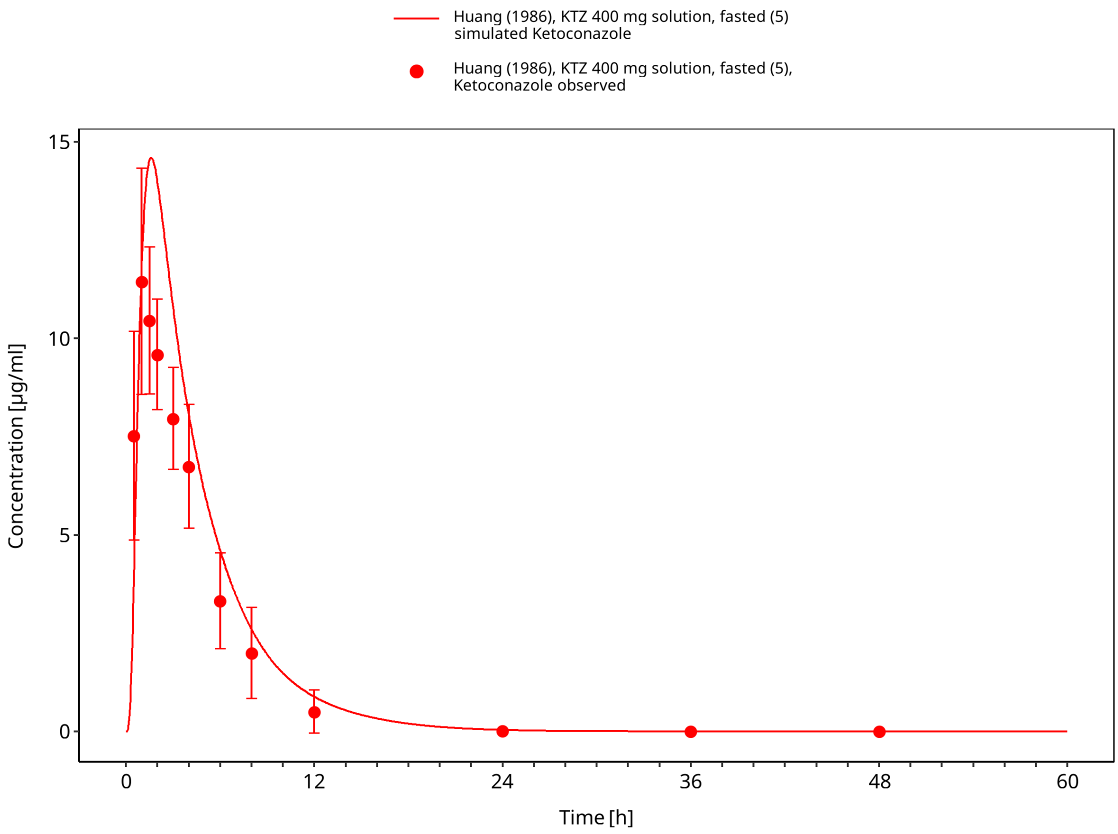


Figure 3-7: Time Profile Analysis

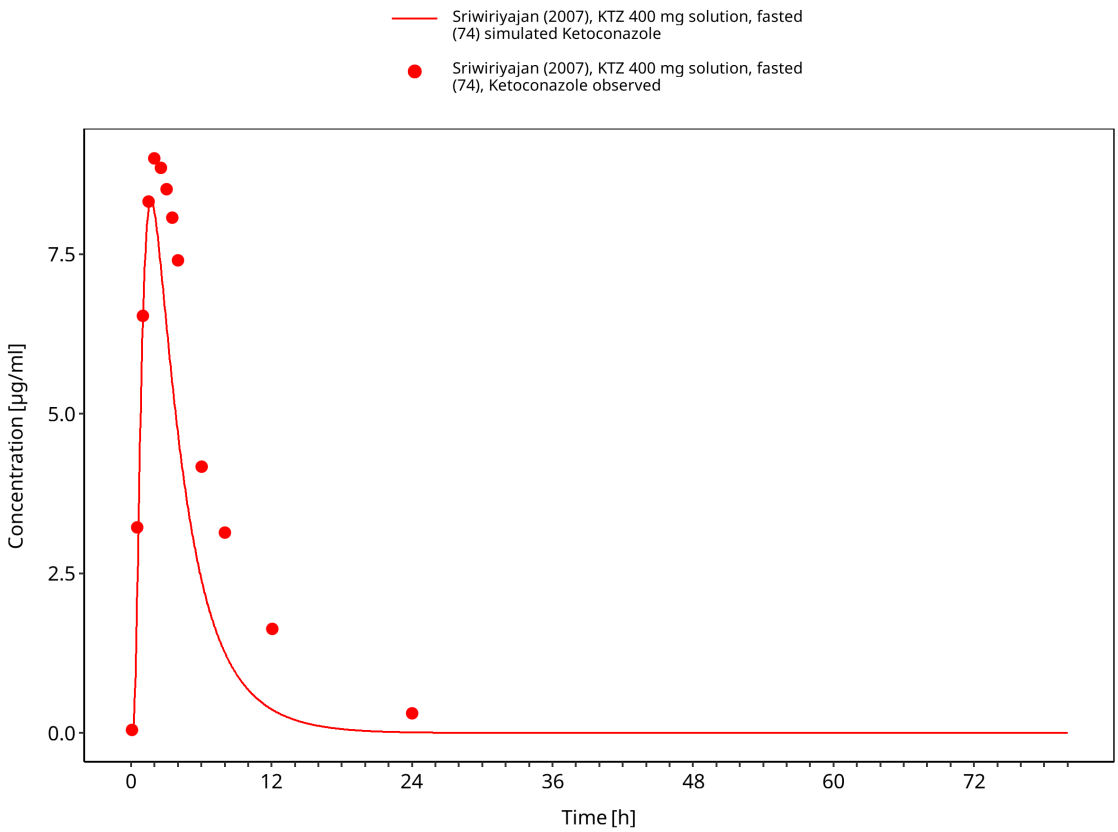


Figure 3-8: Time Profile Analysis

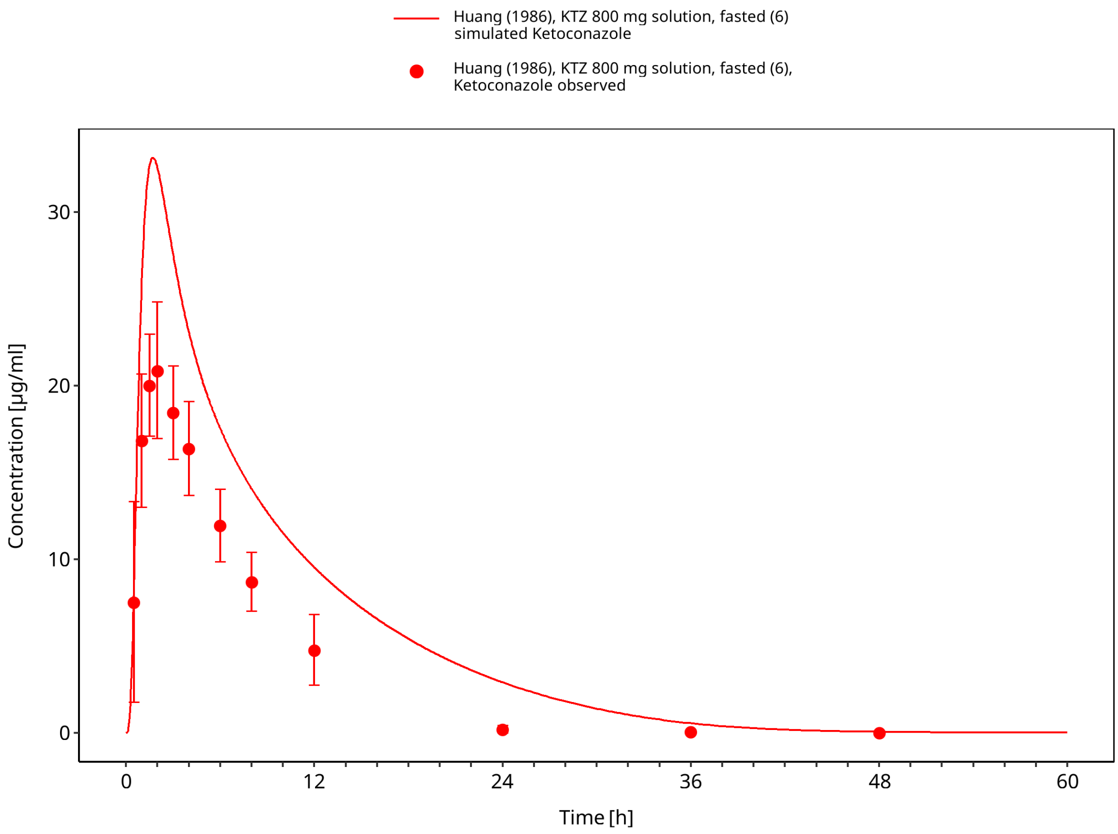


Figure 3-9: Time Profile Analysis

3.3.2 Fasted tablet studies

This section contains fasted oral tablet studies used to evaluate tablet absorption and systemic ketoconazole exposure.

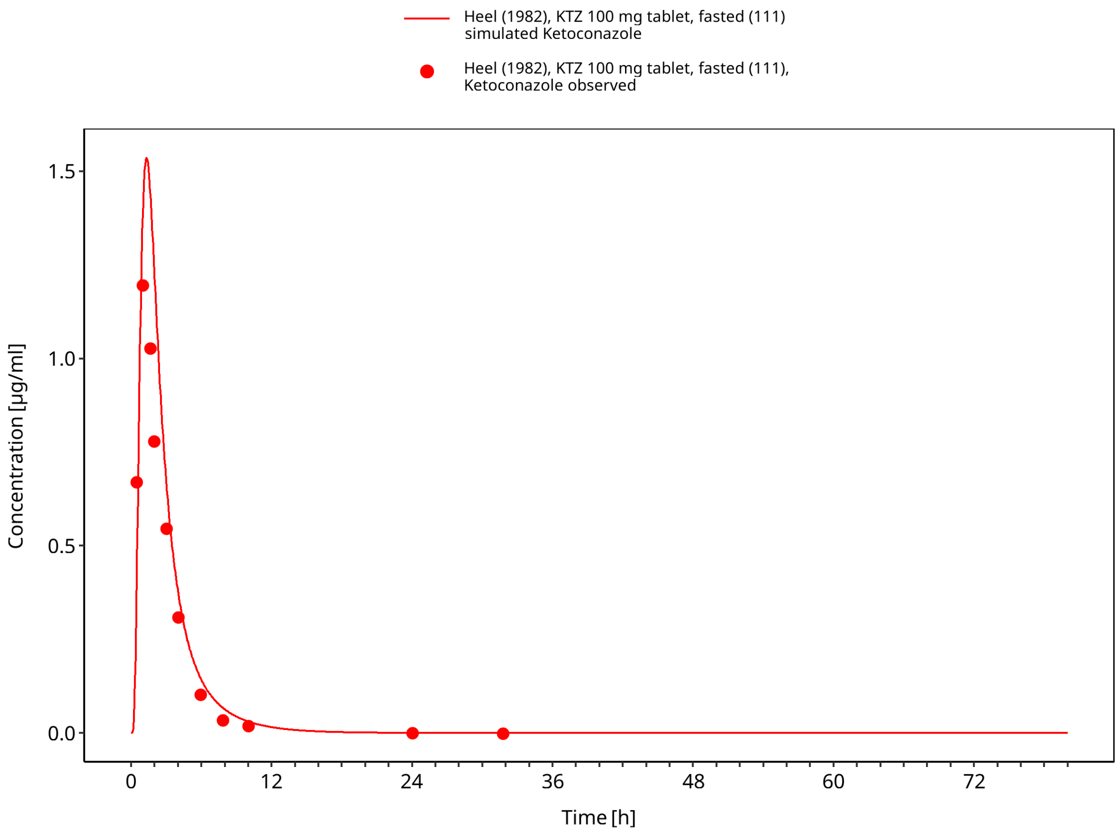


Figure 3-10: Time Profile Analysis

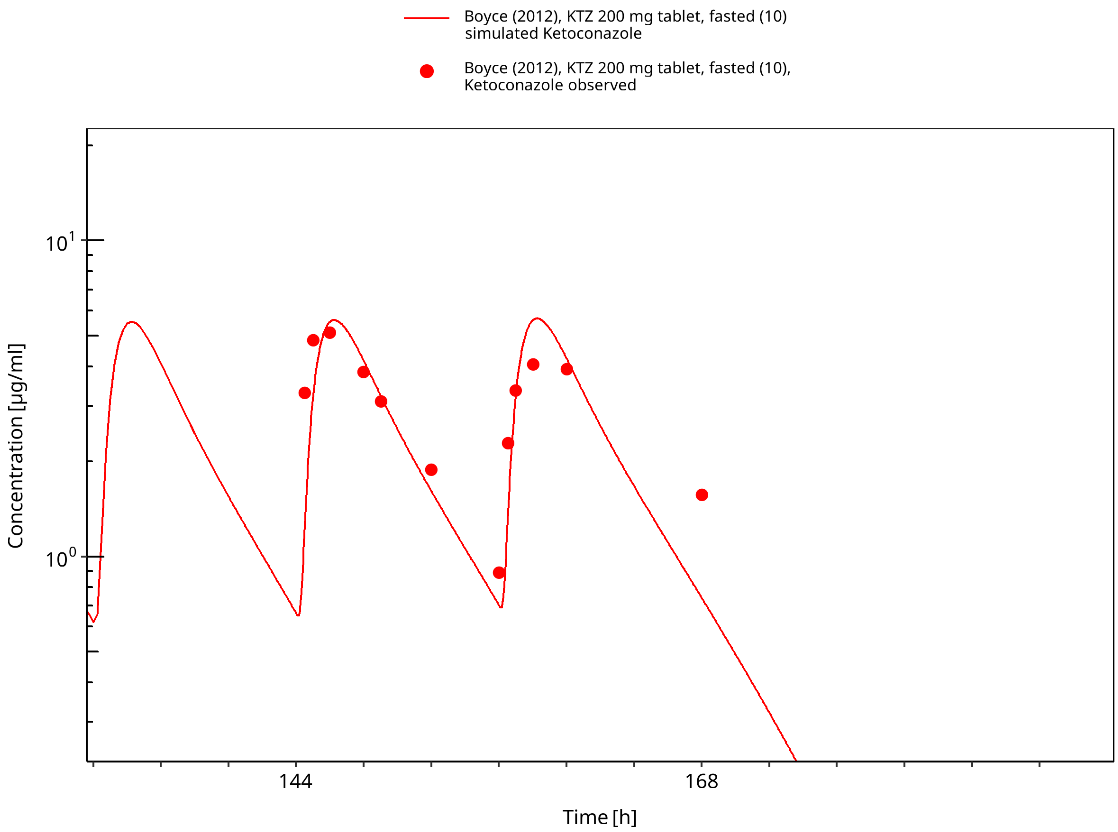


Figure 3-11: Time Profile Analysis

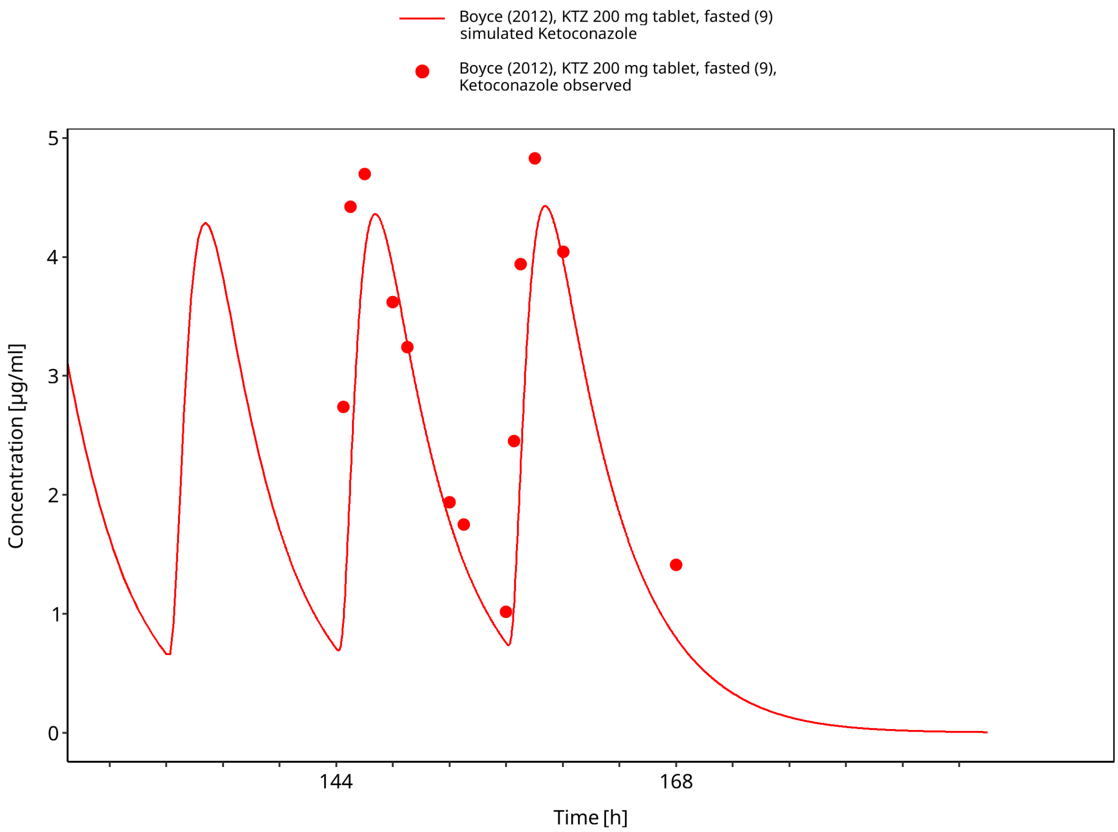


Figure 3-12: Time Profile Analysis

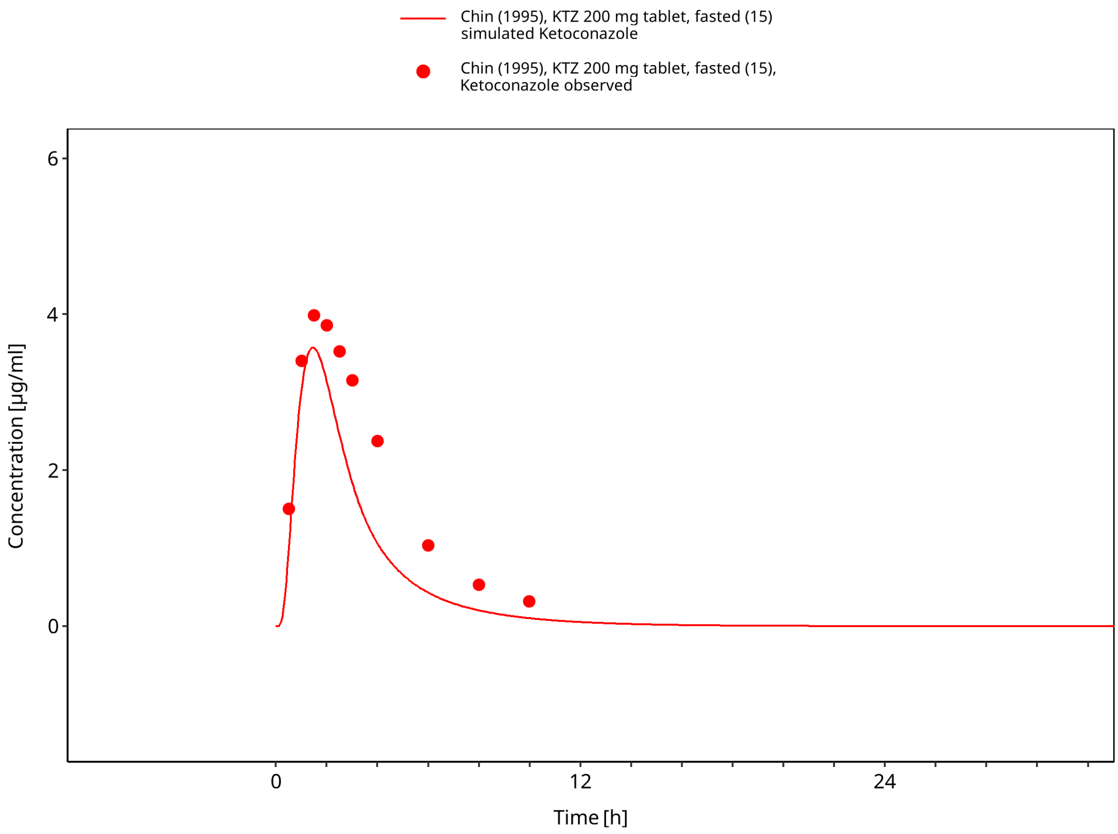
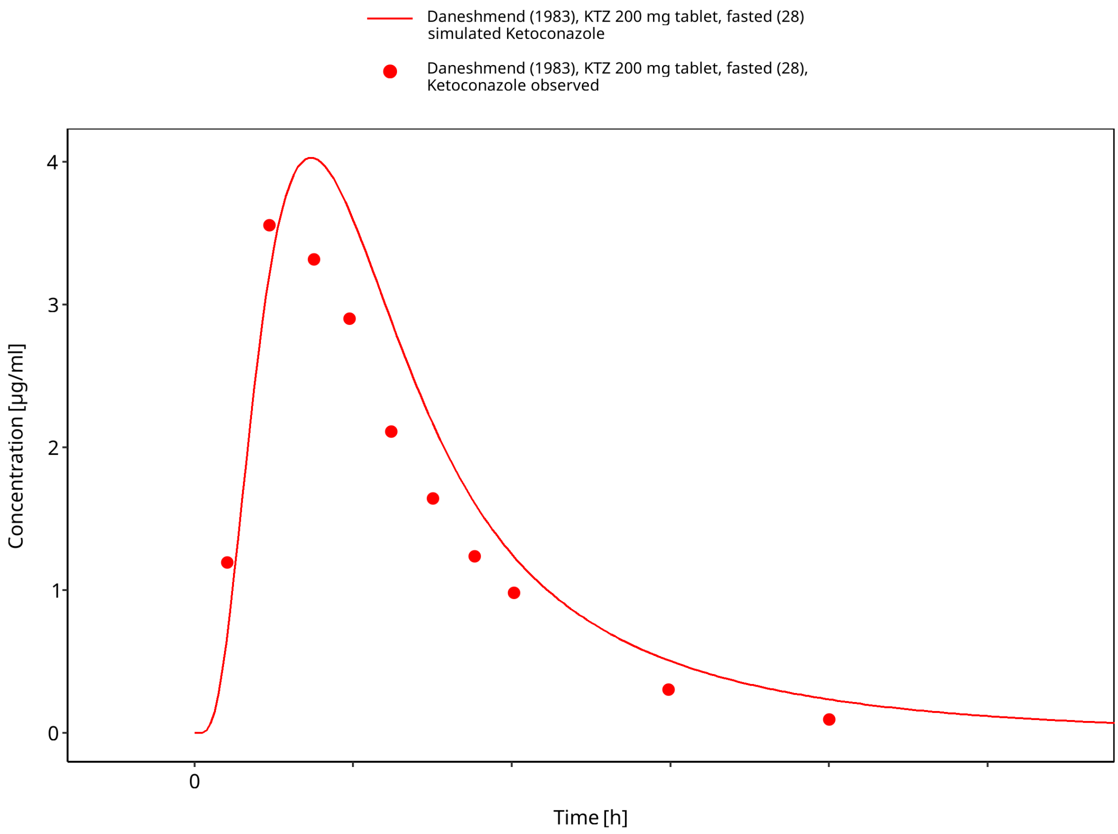


Figure 3-13: Time Profile Analysis



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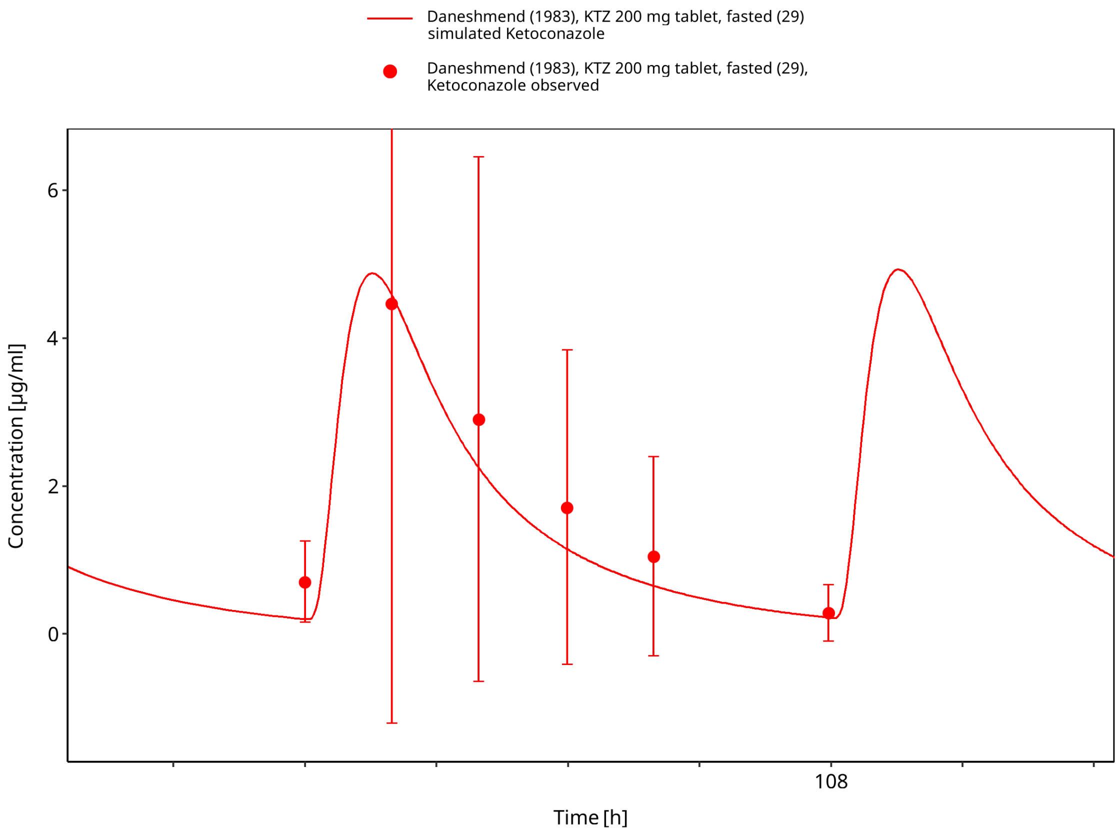
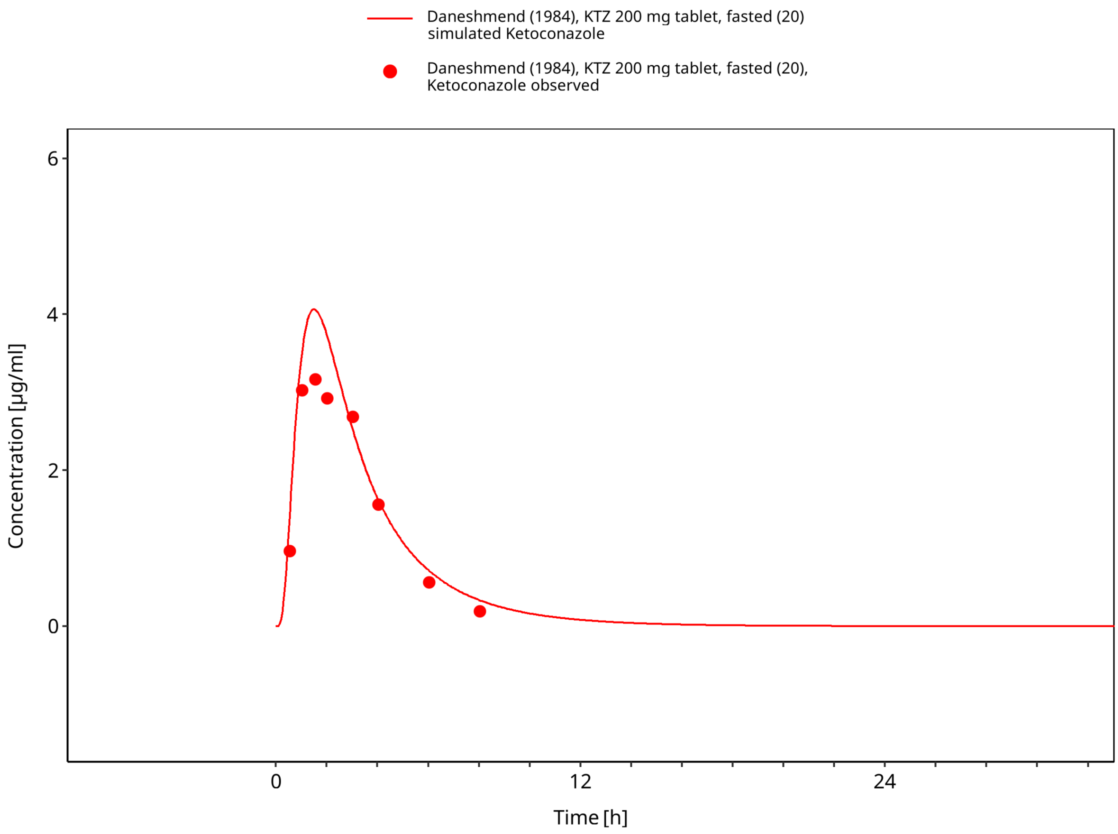


Figure 3-15: Time Profile Analysis



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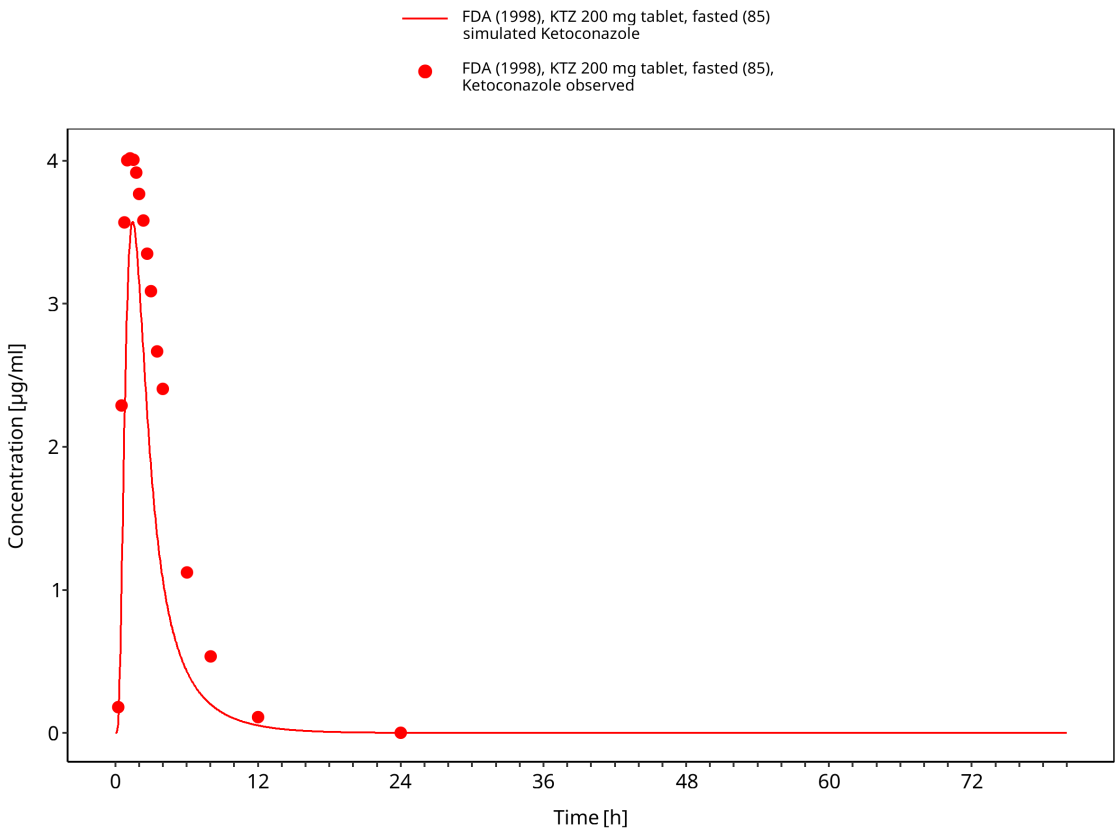

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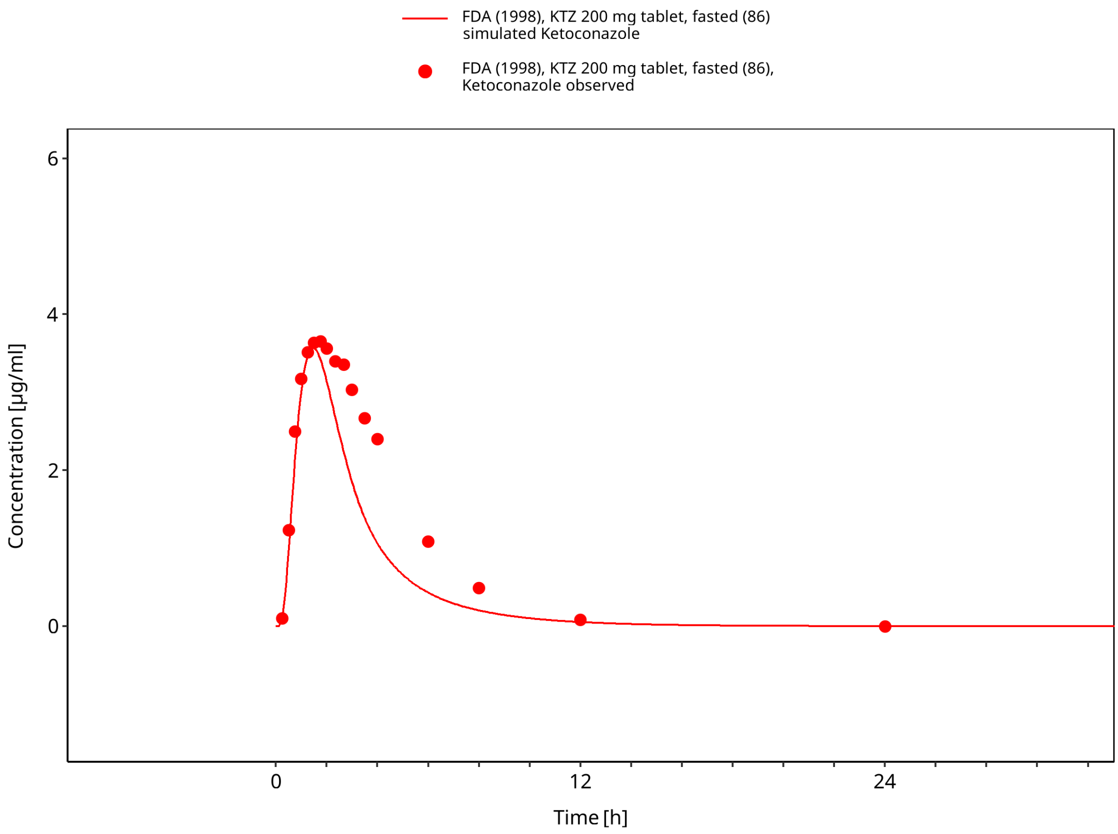

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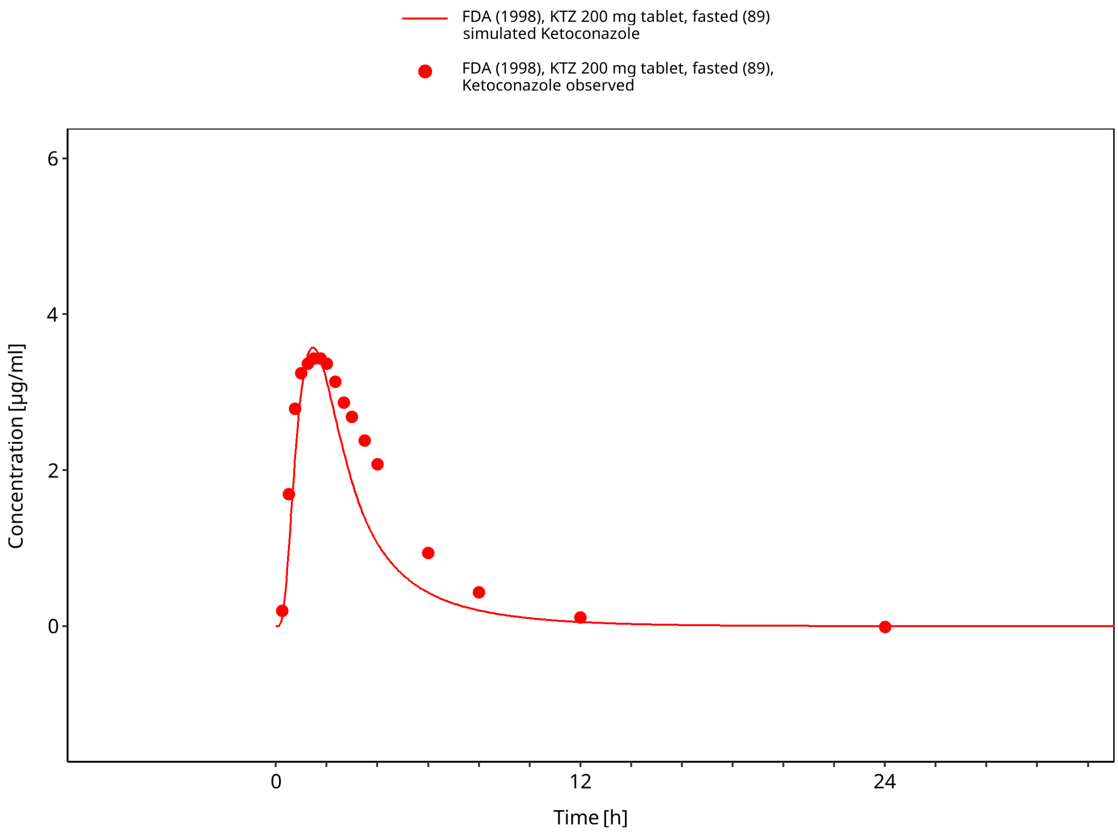

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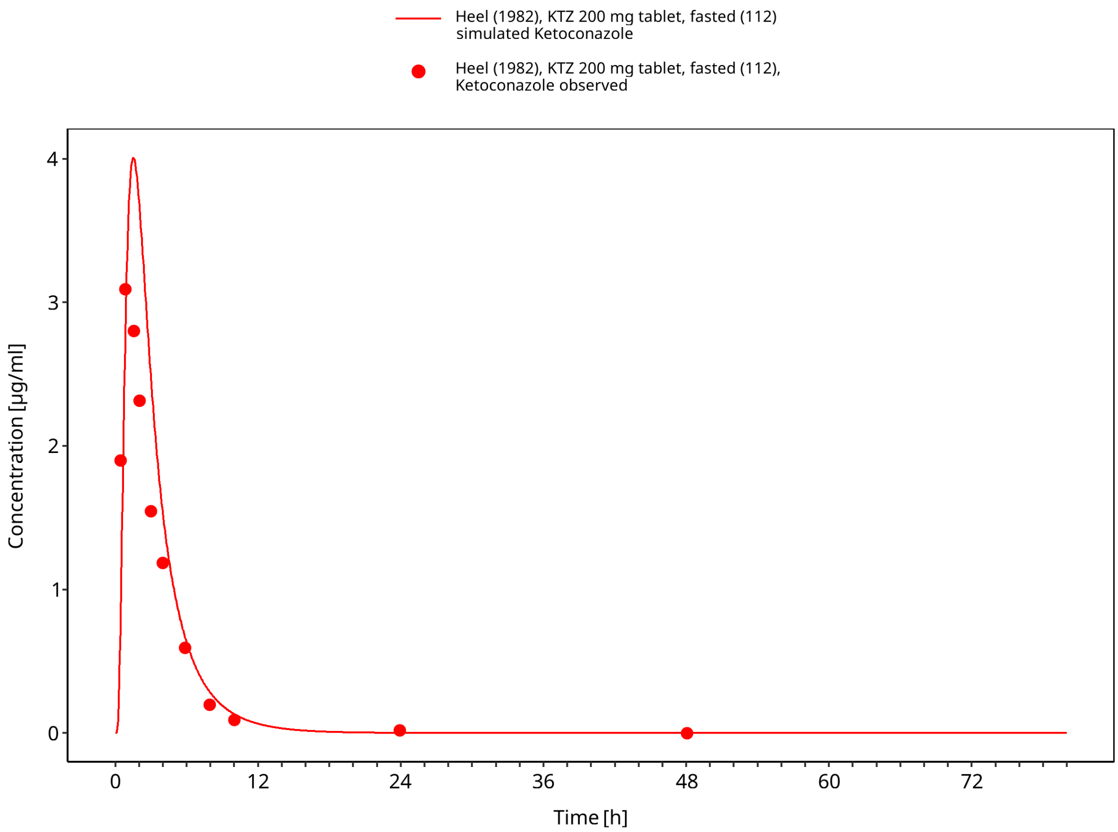

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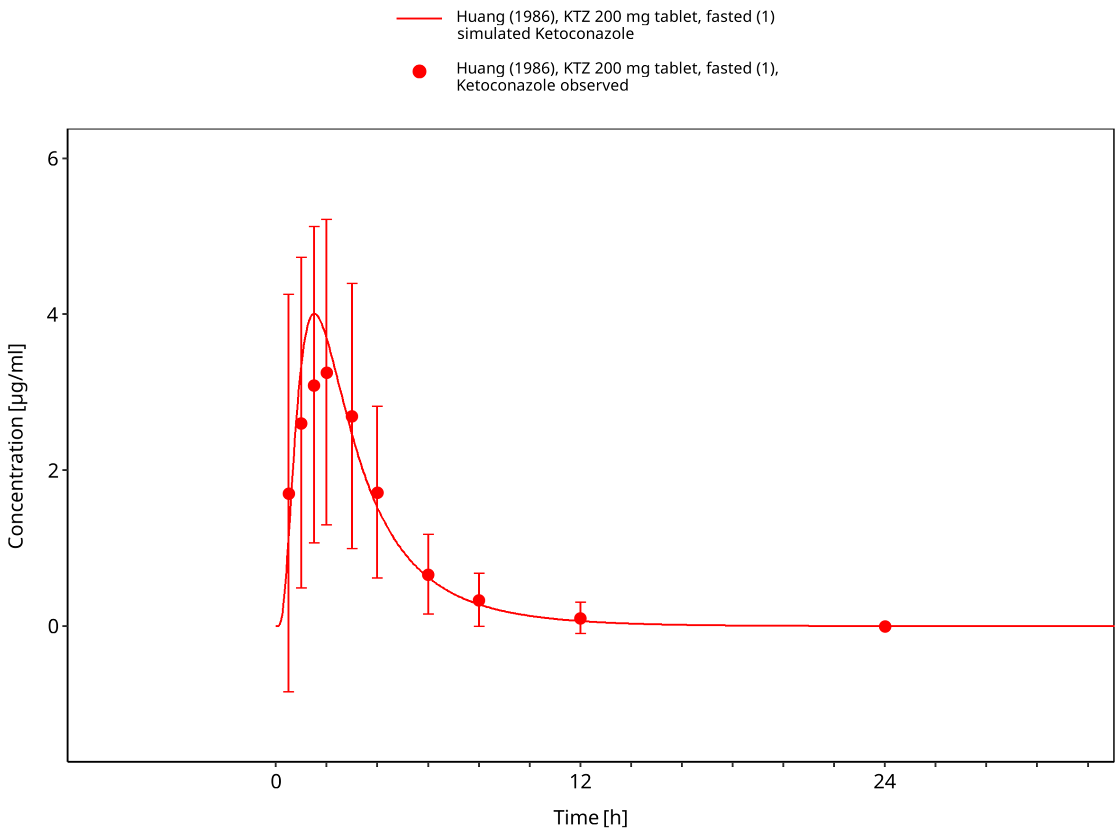

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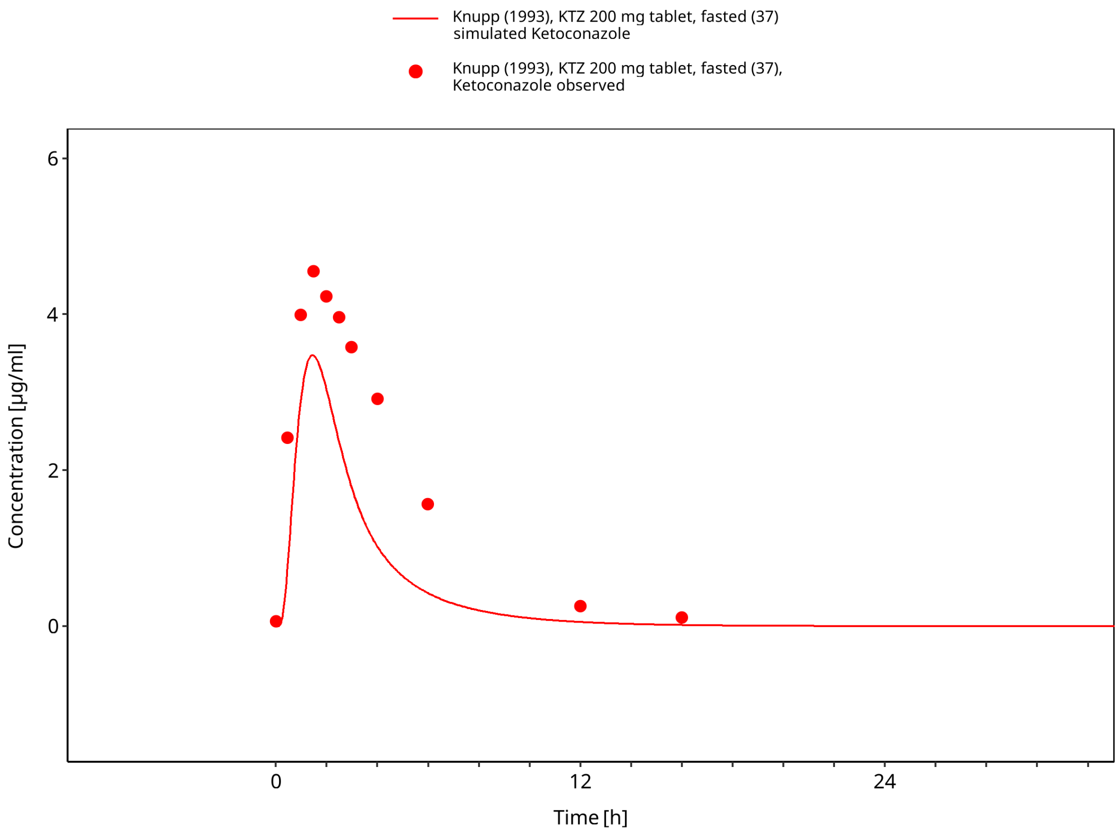

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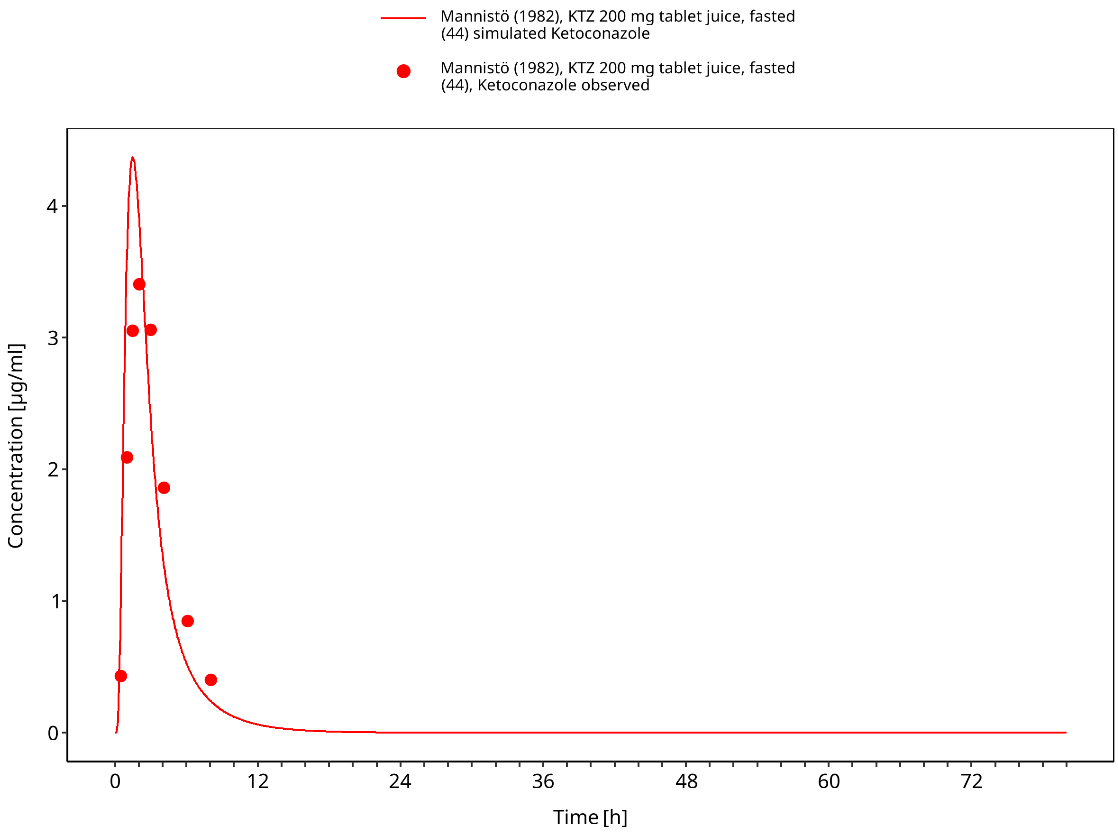

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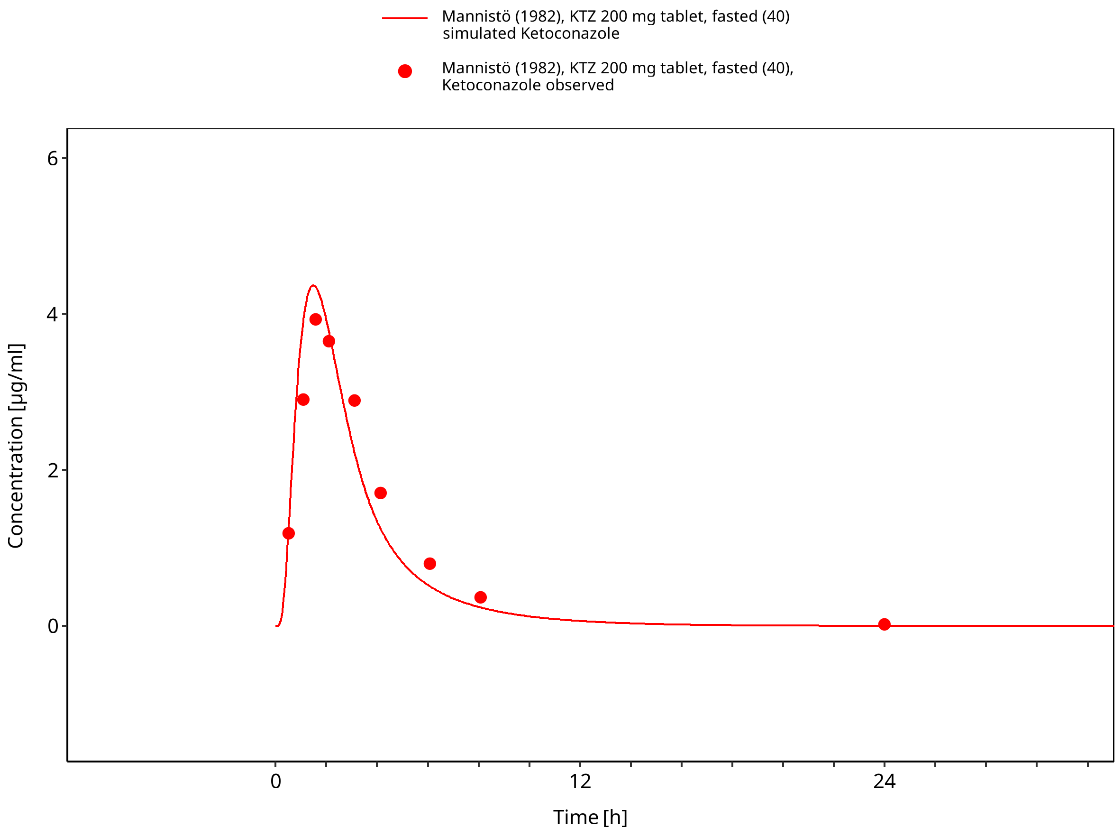

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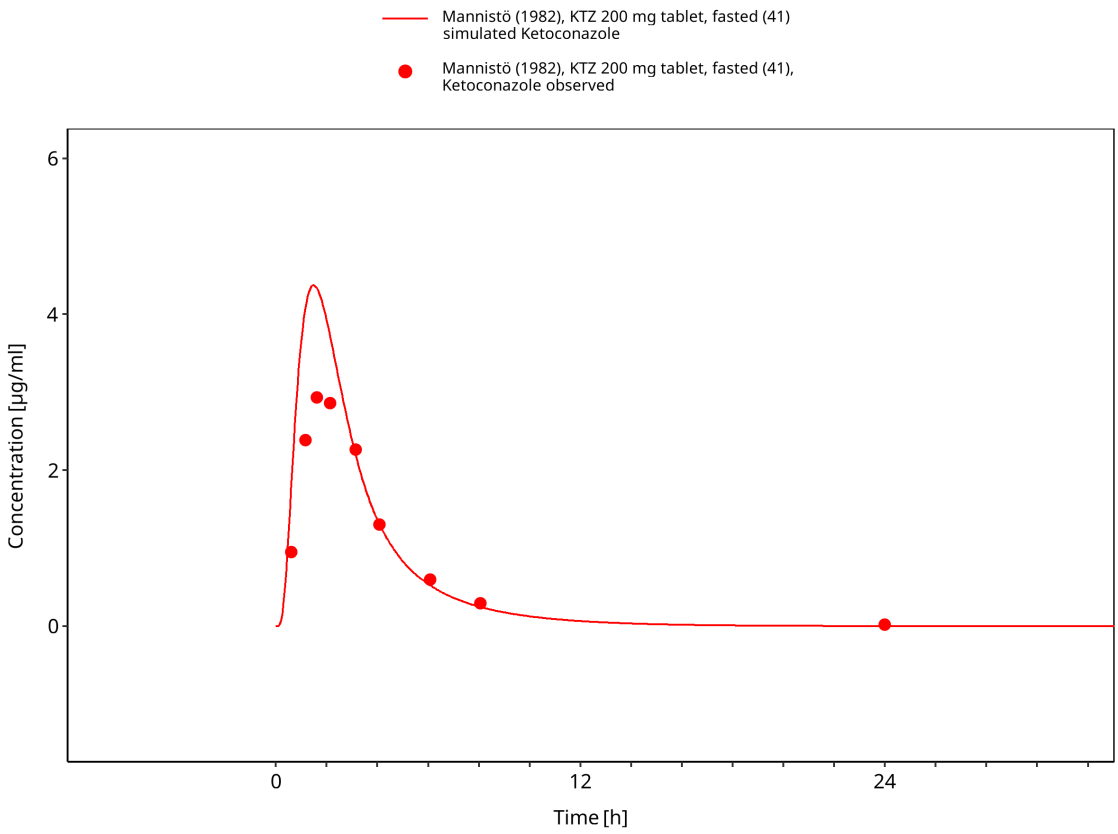

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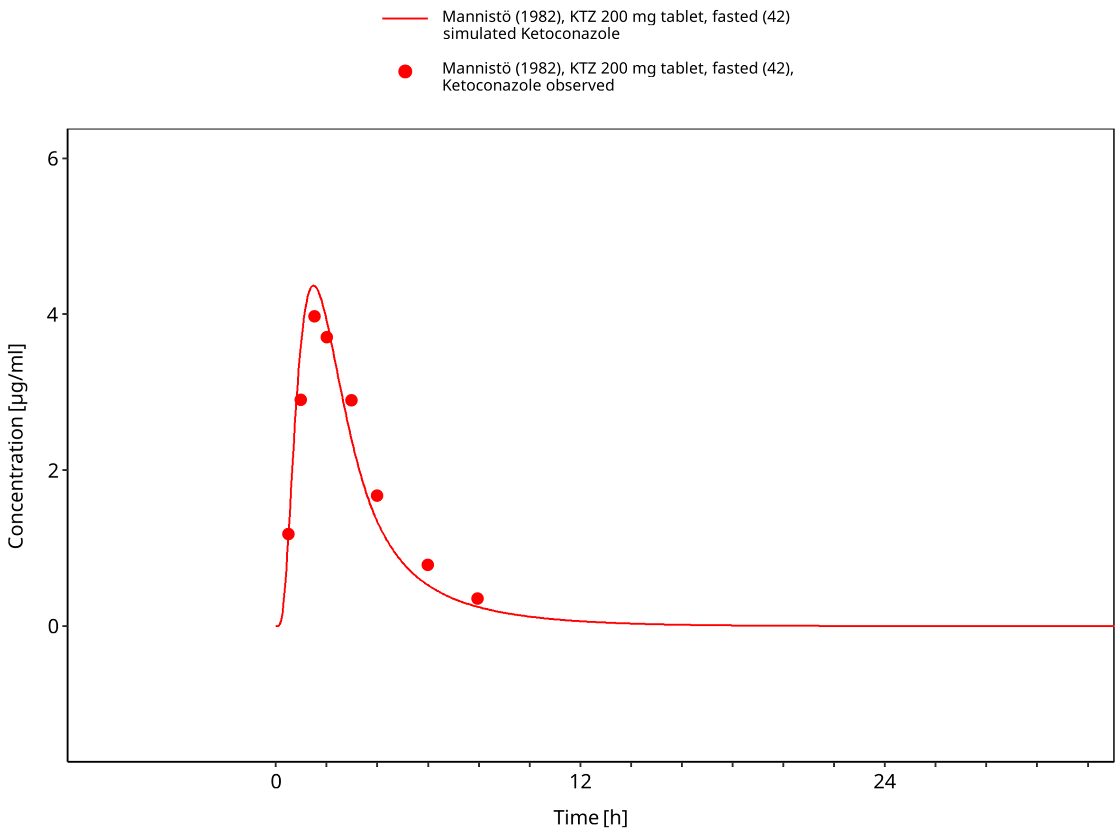

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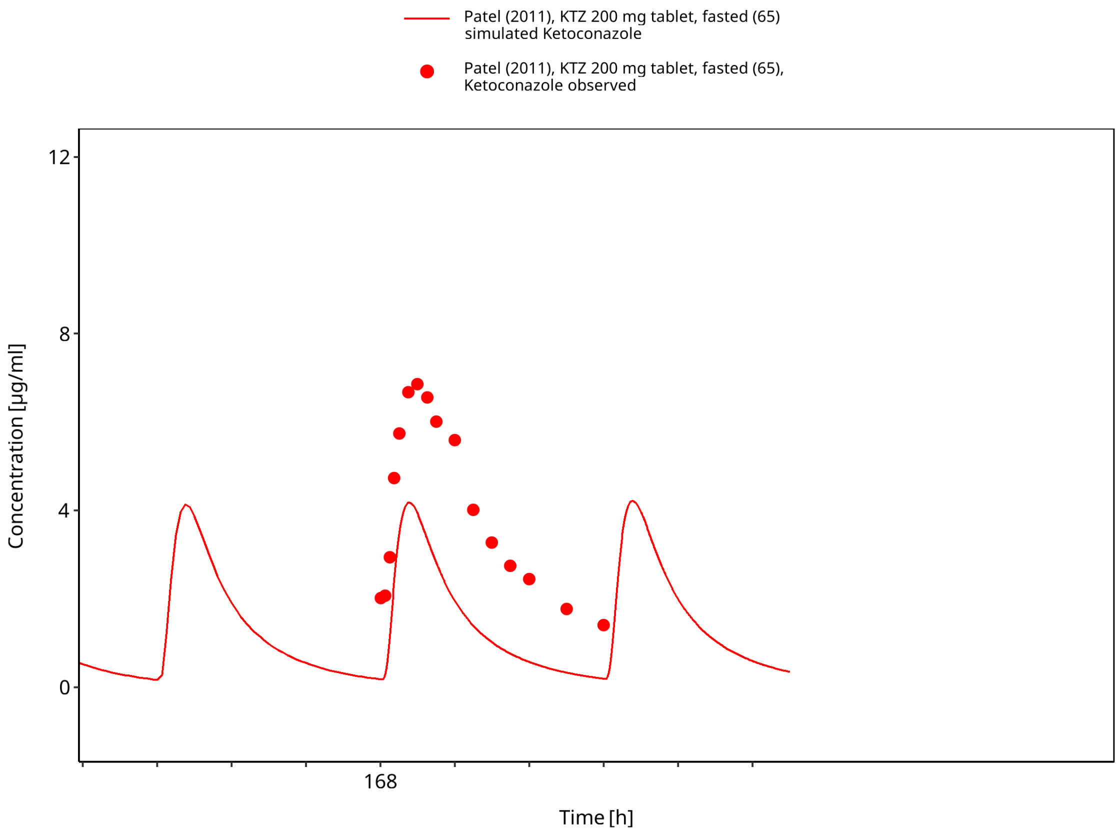

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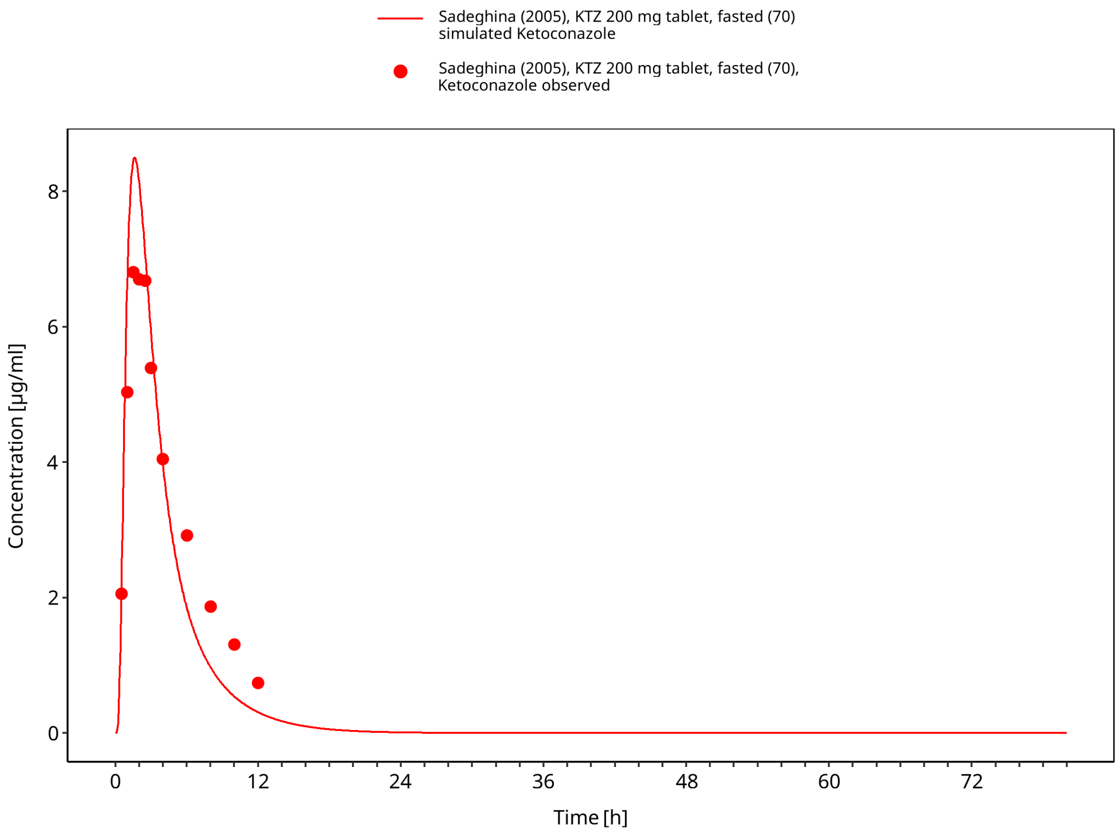

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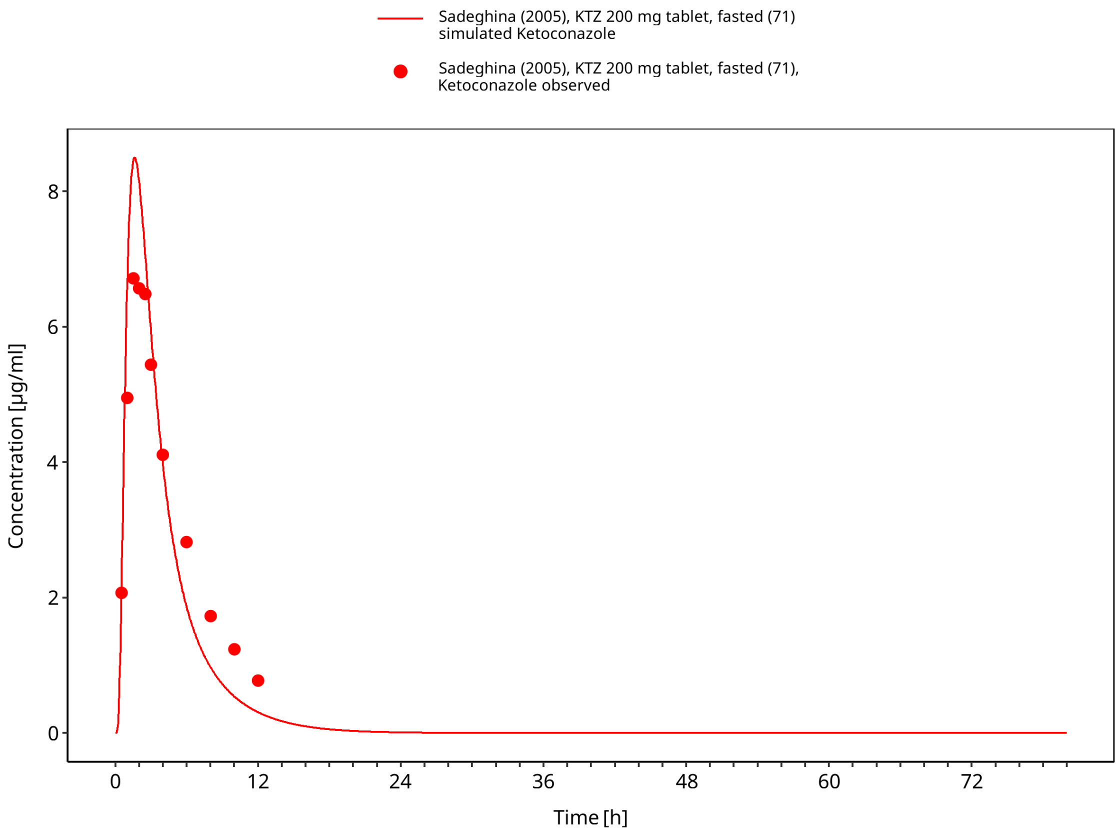

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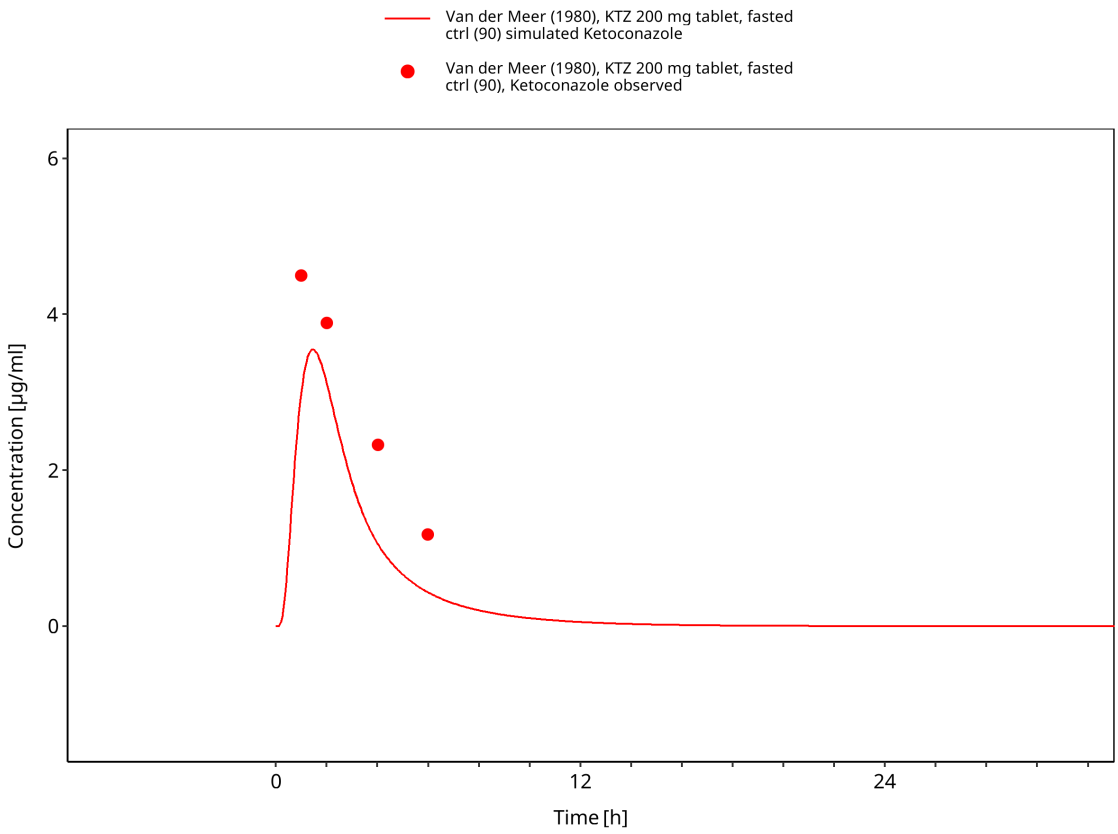

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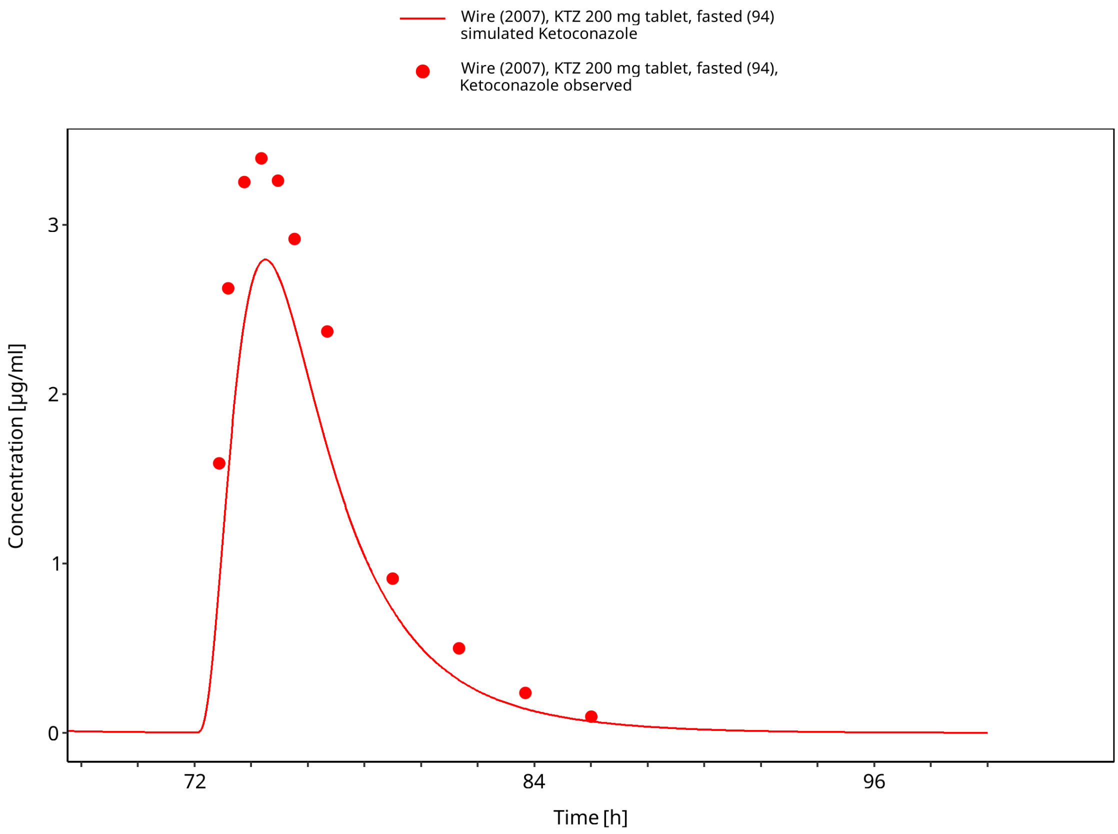

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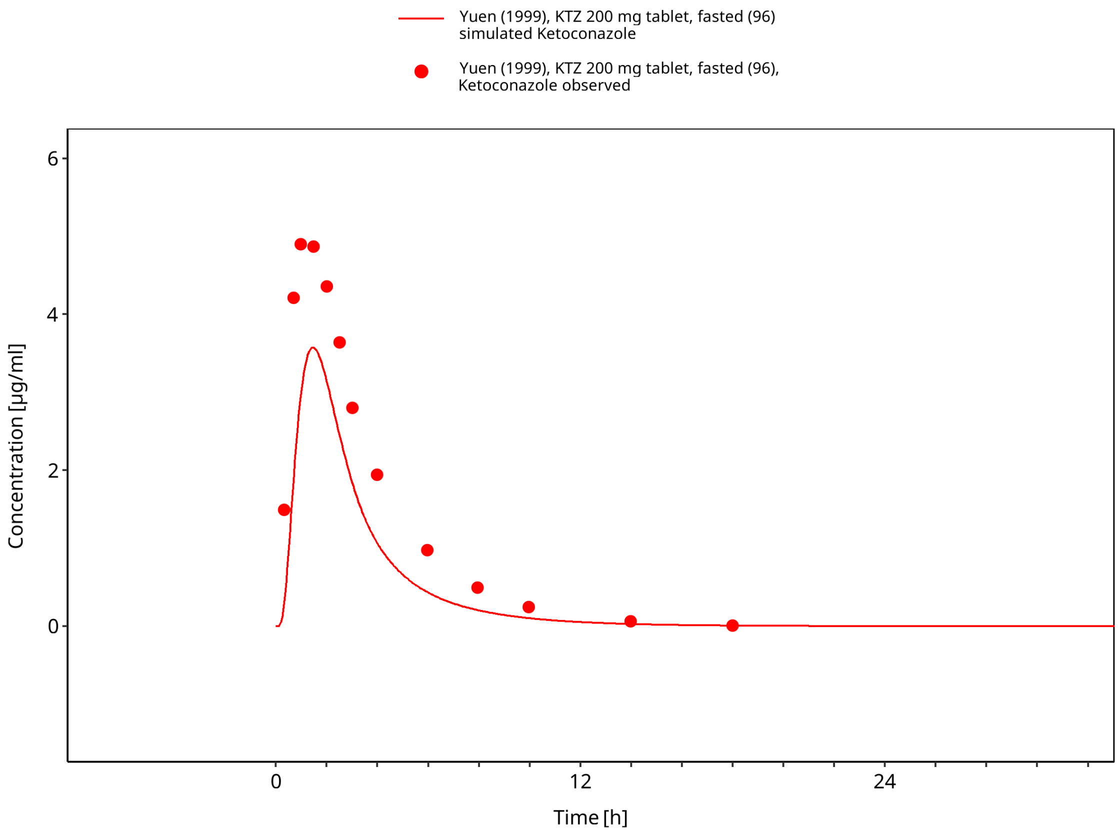

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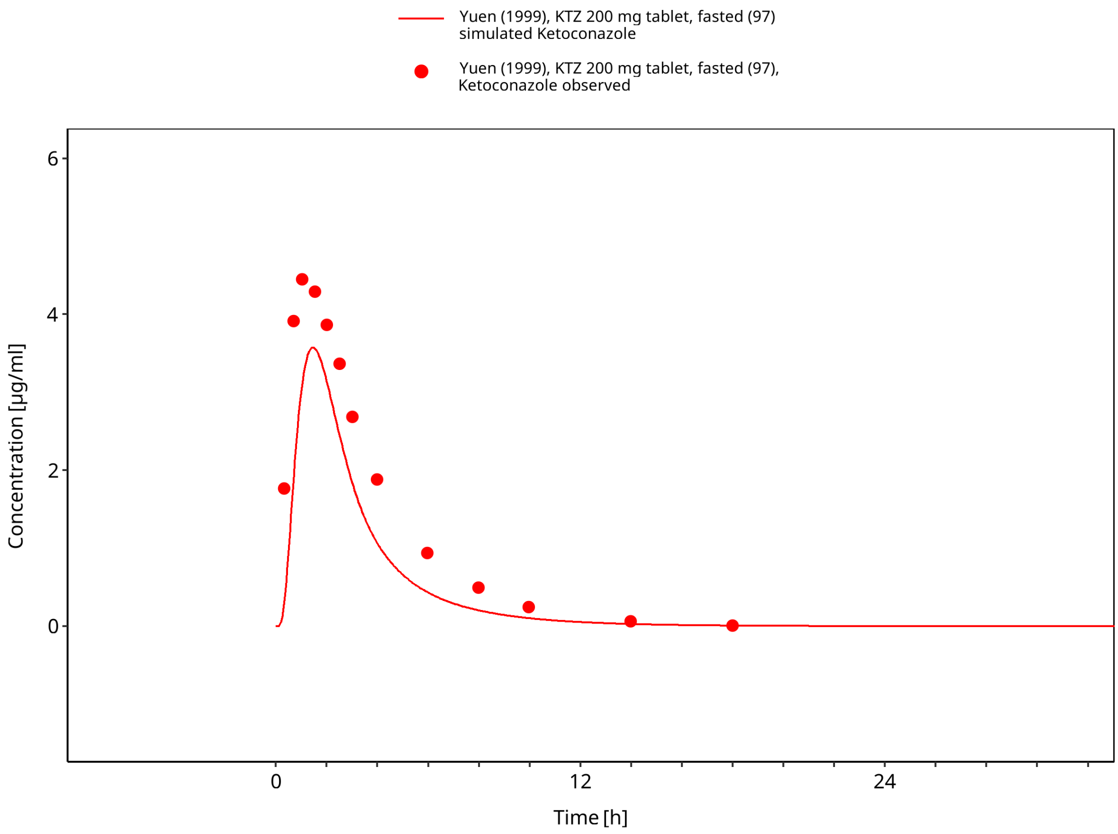

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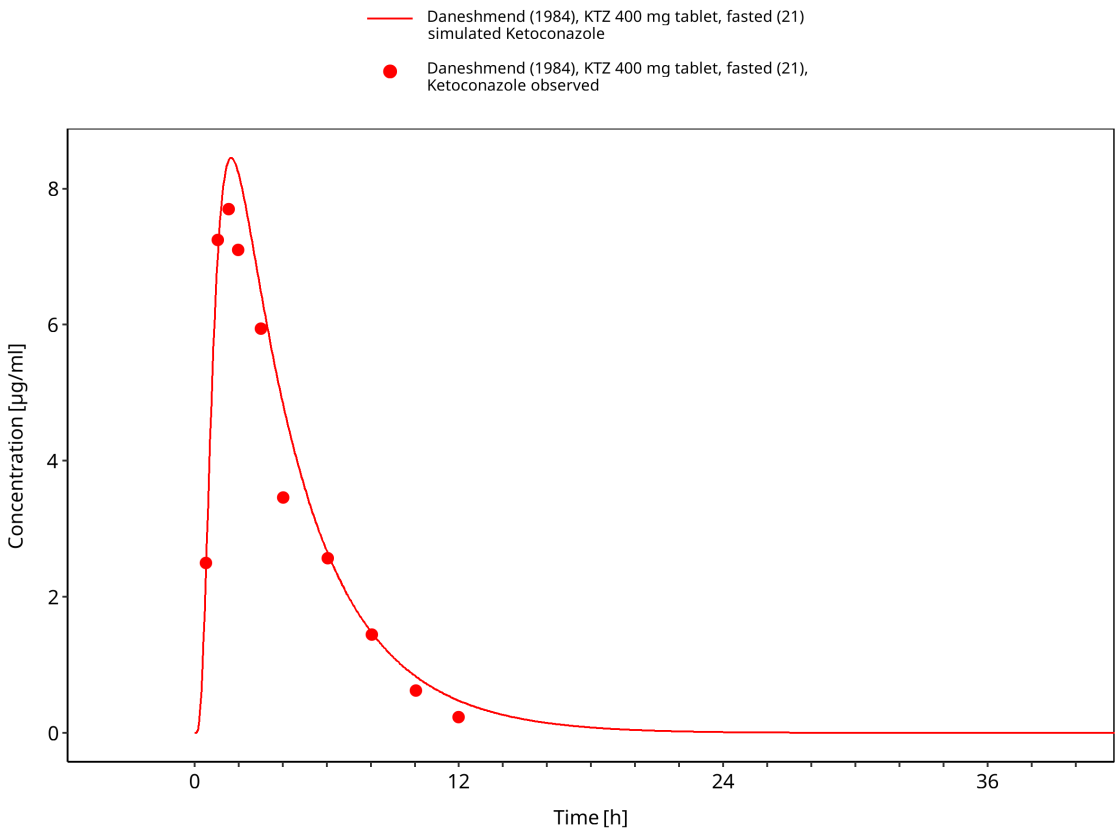

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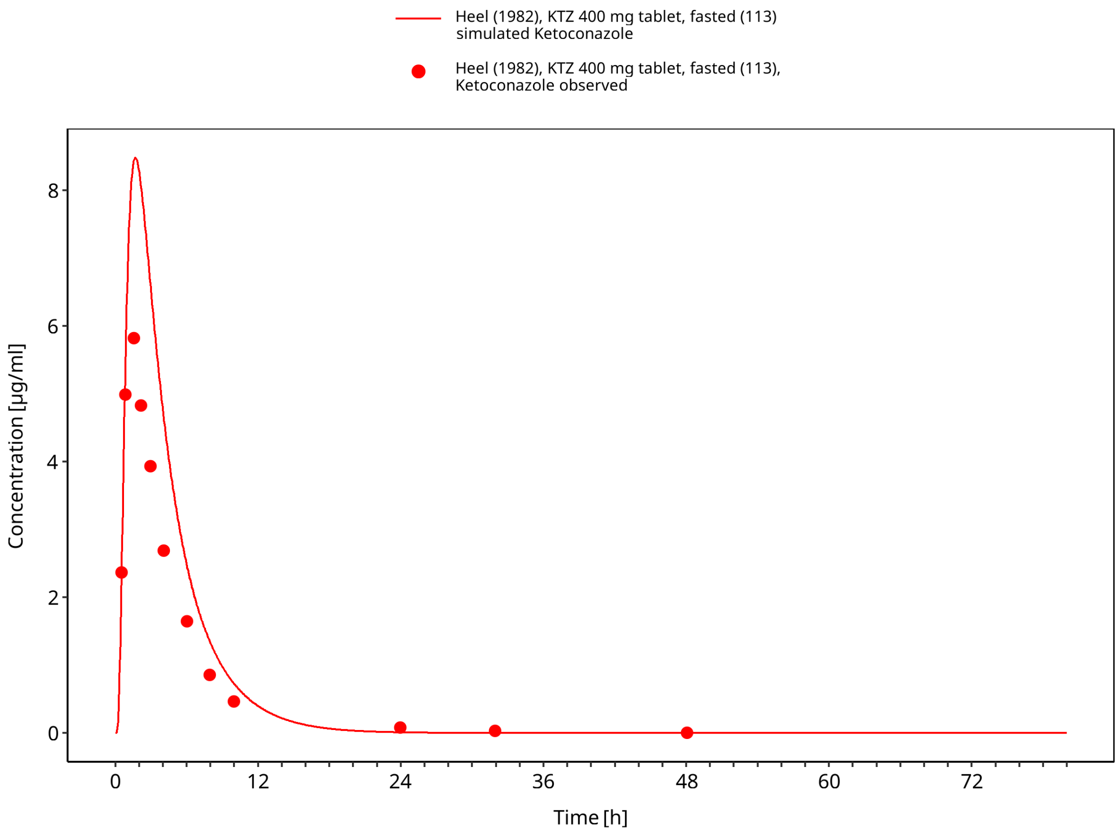

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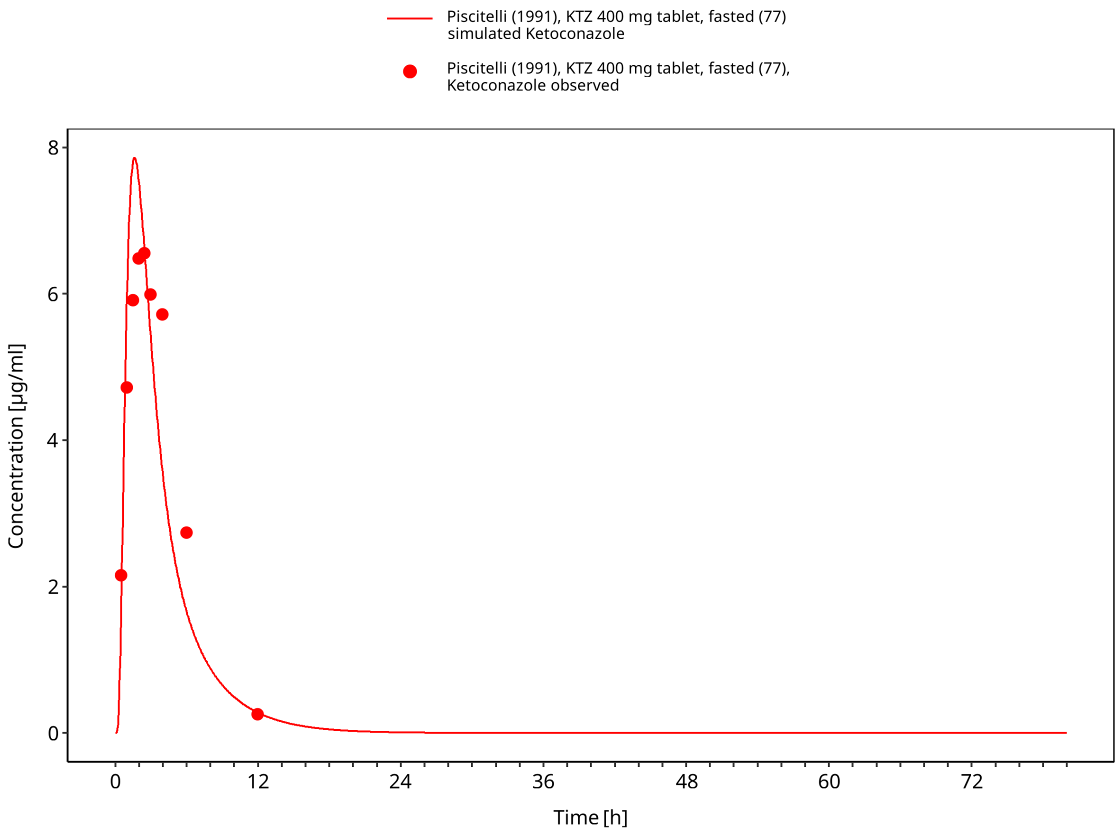

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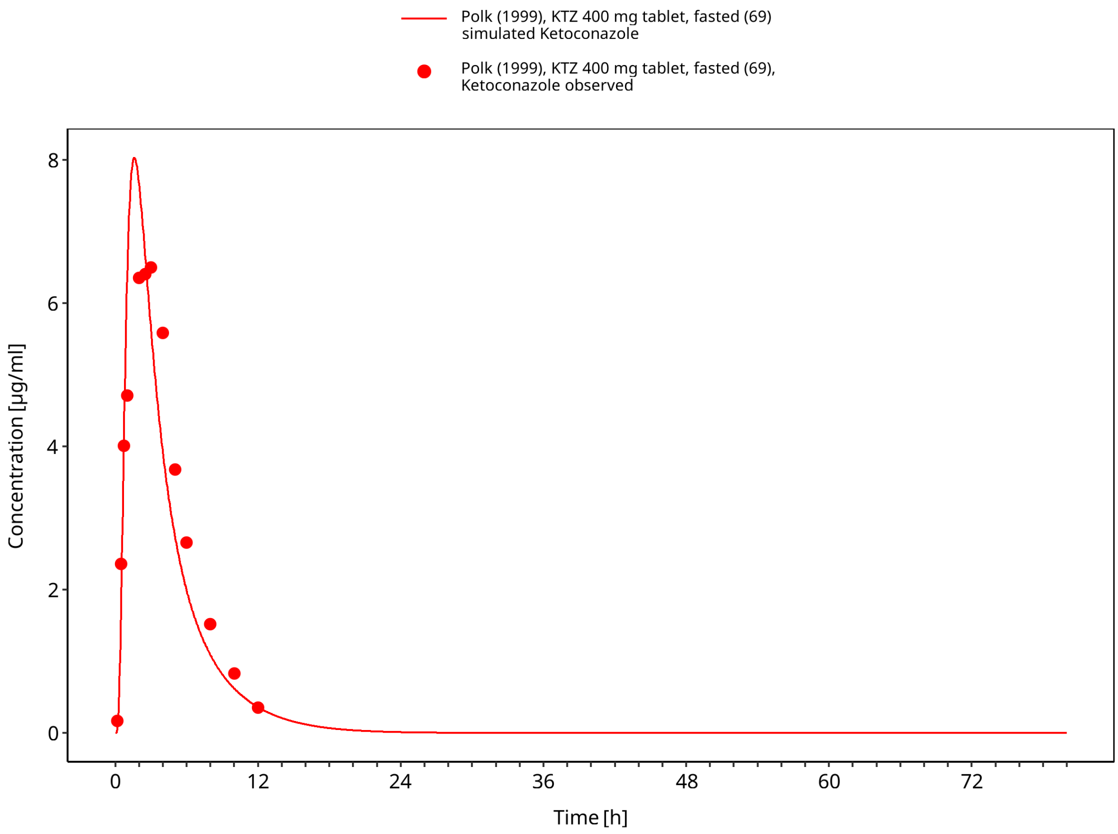

Figure 3-37: Time Profile Analysis

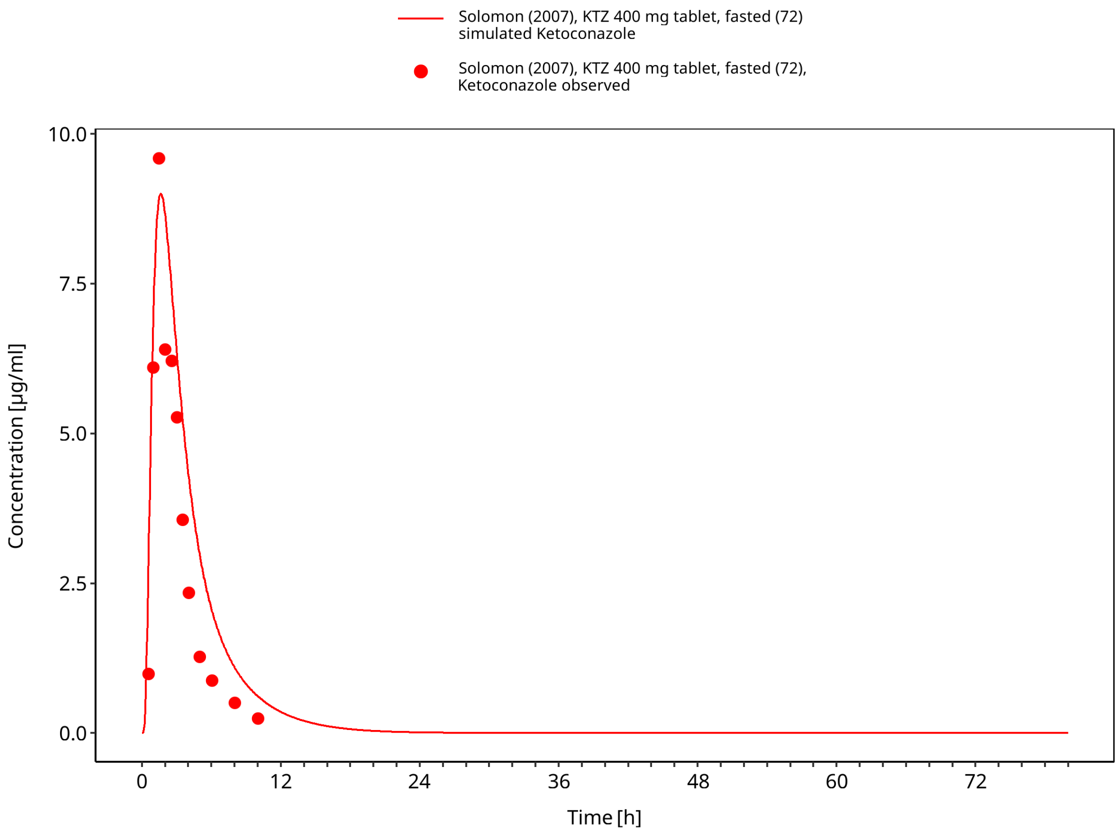

Figure 3-38: Time Profile Analysis

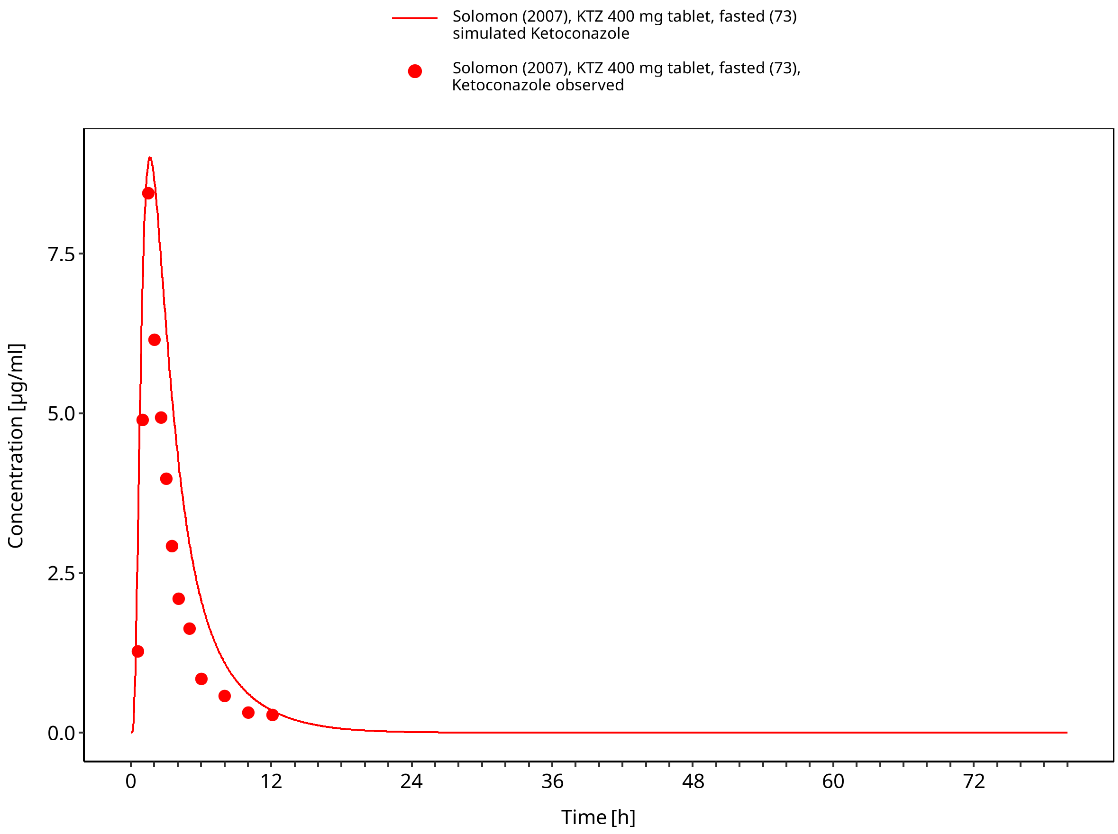

Figure 3-39: Time Profile Analysis

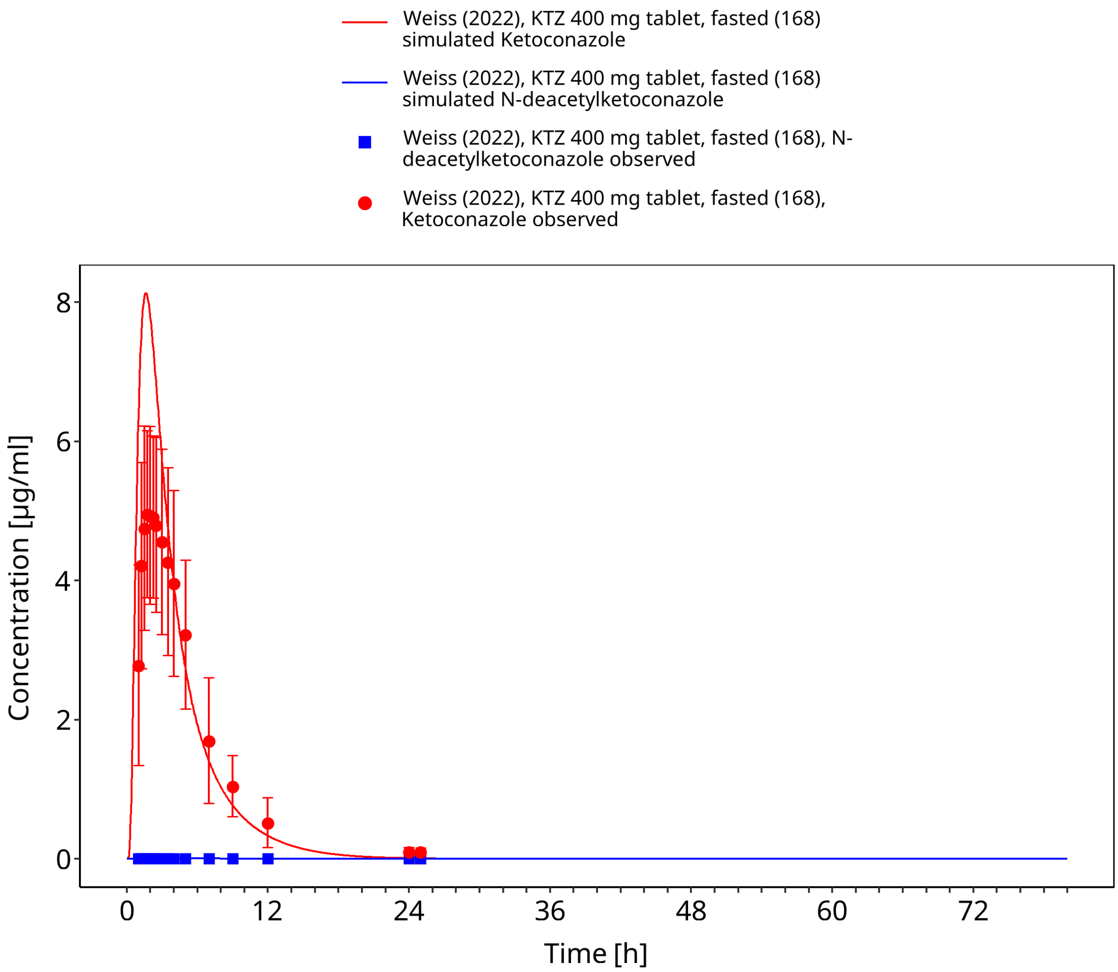

Figure 3-40: Time Profile Analysis

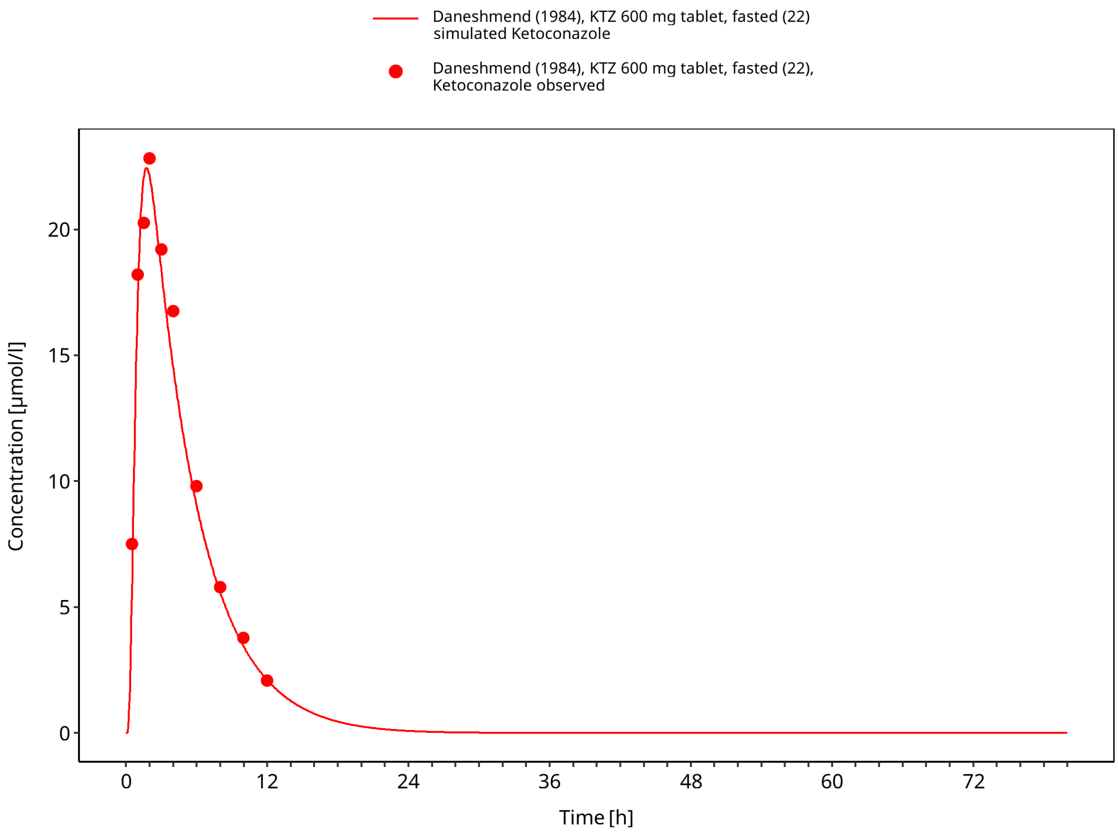

Figure 3-41: Time Profile Analysis

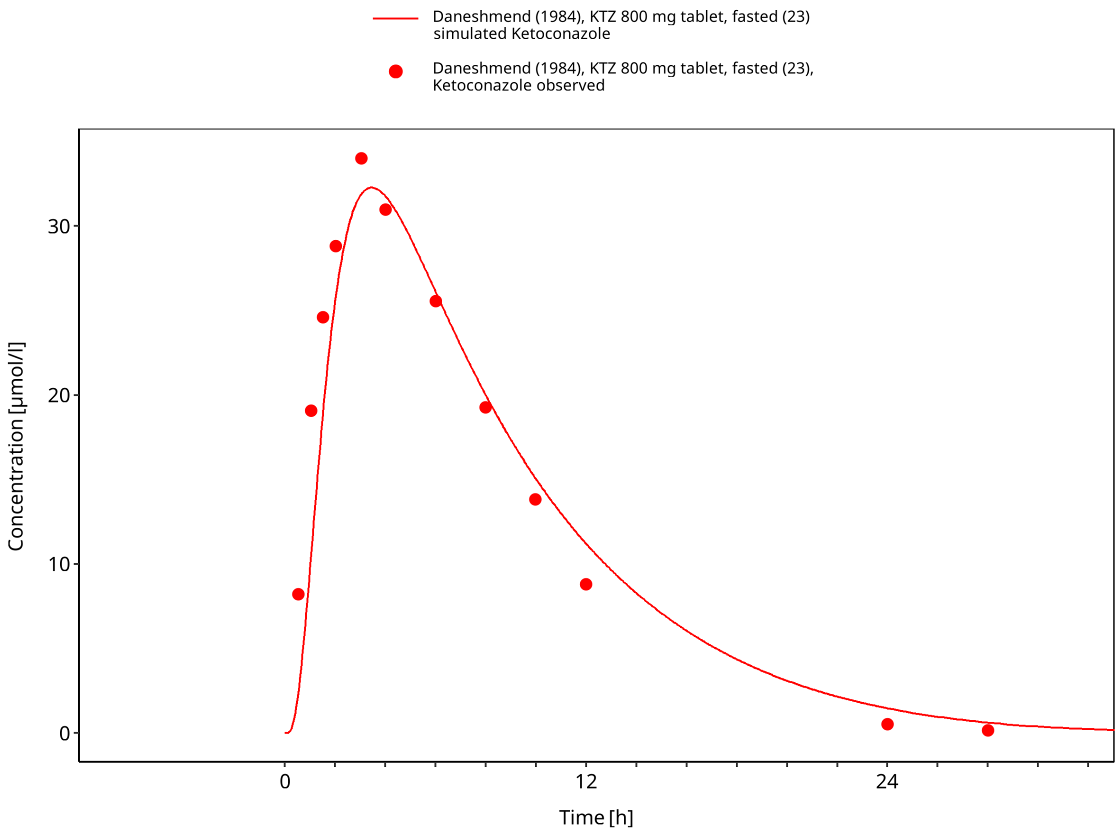

Figure 3-42: Time Profile Analysis

3.3.3 Fed tablet and food-effect studies

This section contains fed tablet and food-effect studies used to evaluate the influence of gastrointestinal conditions on ketoconazole absorption.

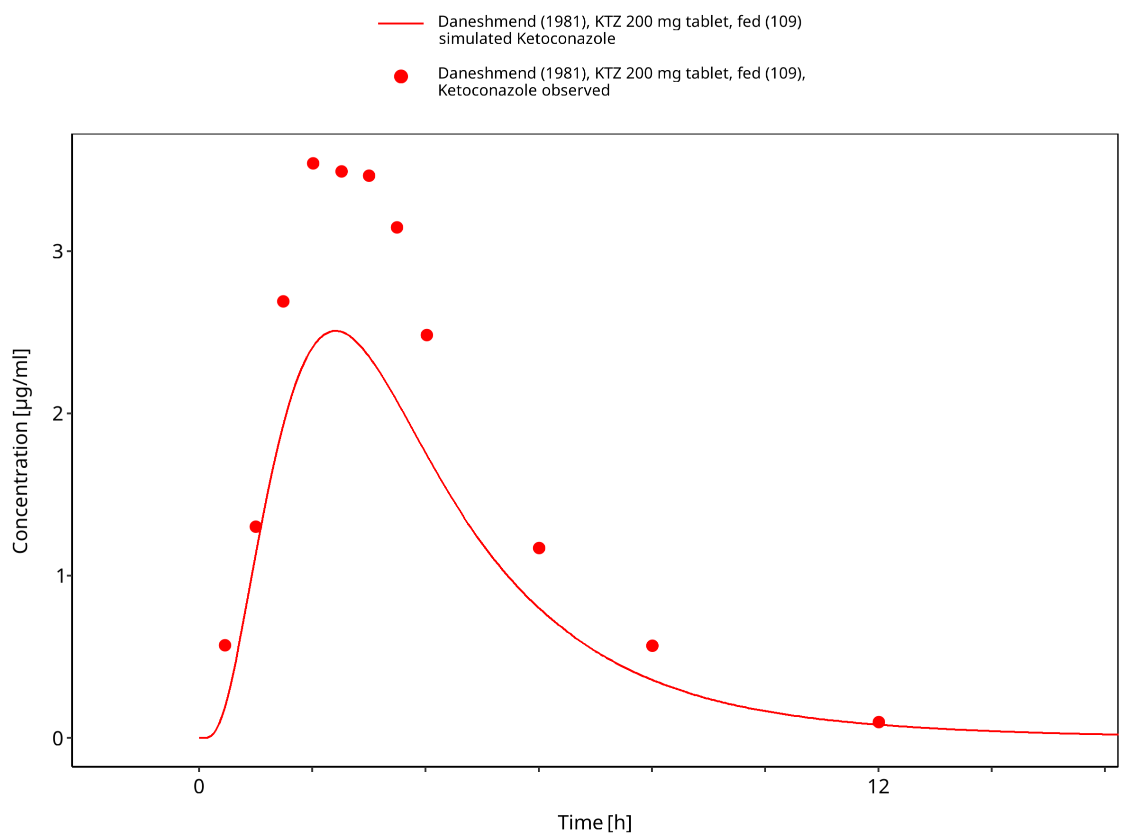

Figure 3-43: Time Profile Analysis

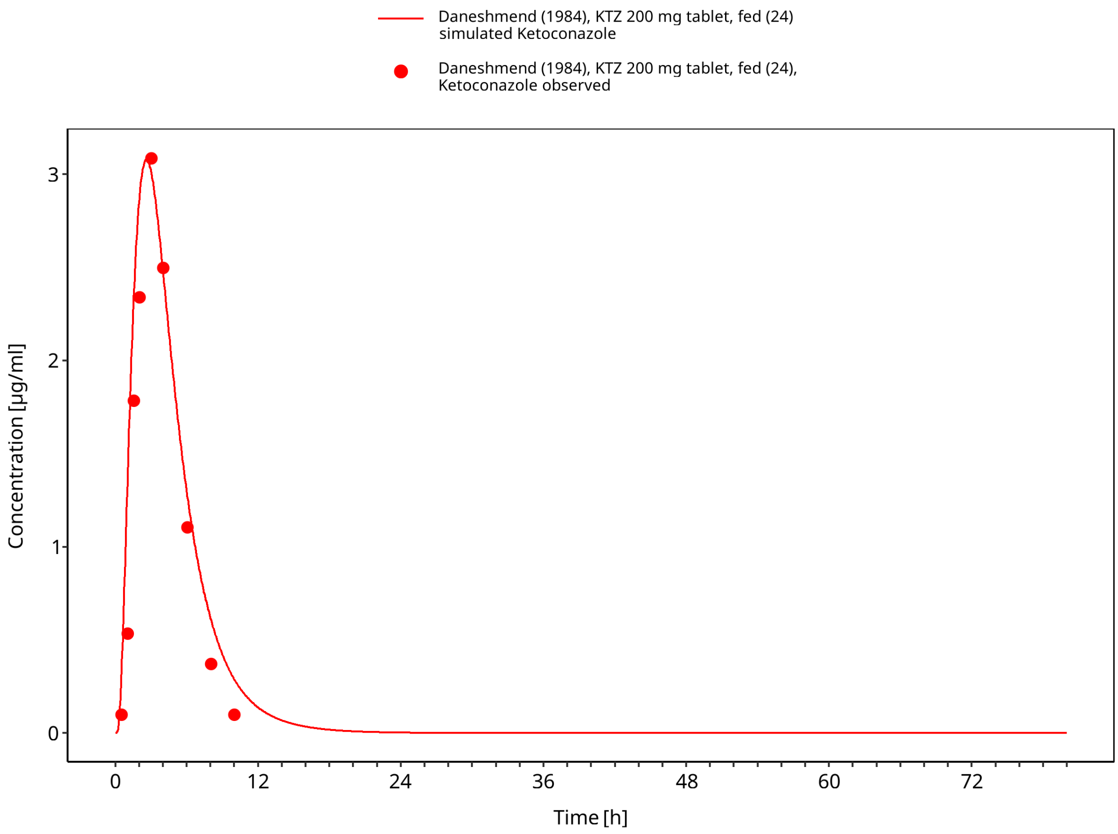

Figure 3-44: Time Profile Analysis

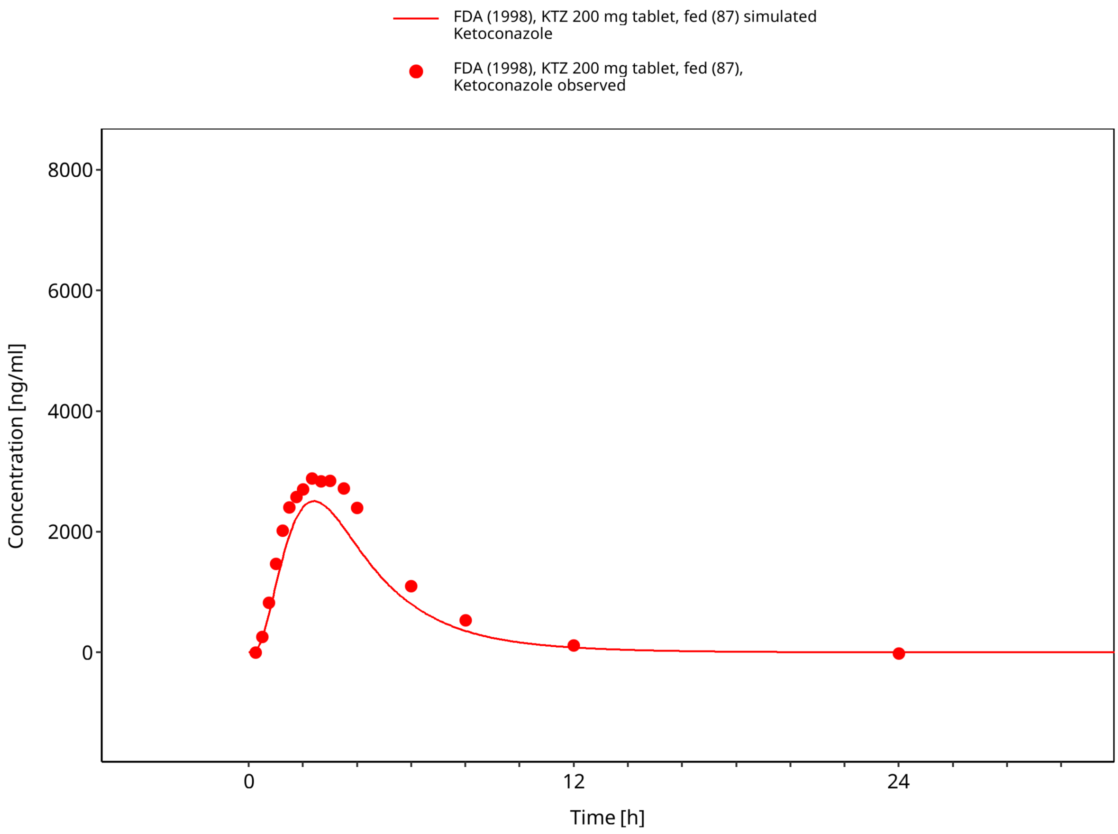

Figure 3-45: Time Profile Analysis

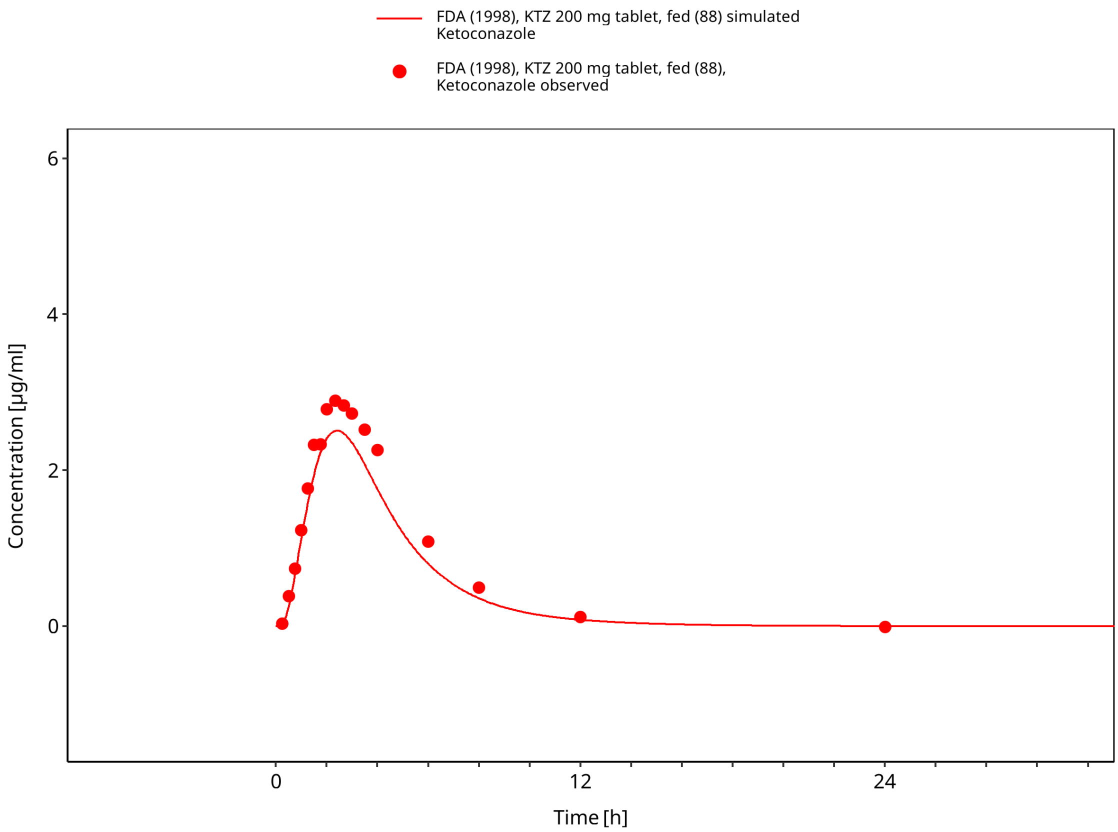

Figure 3-46: Time Profile Analysis

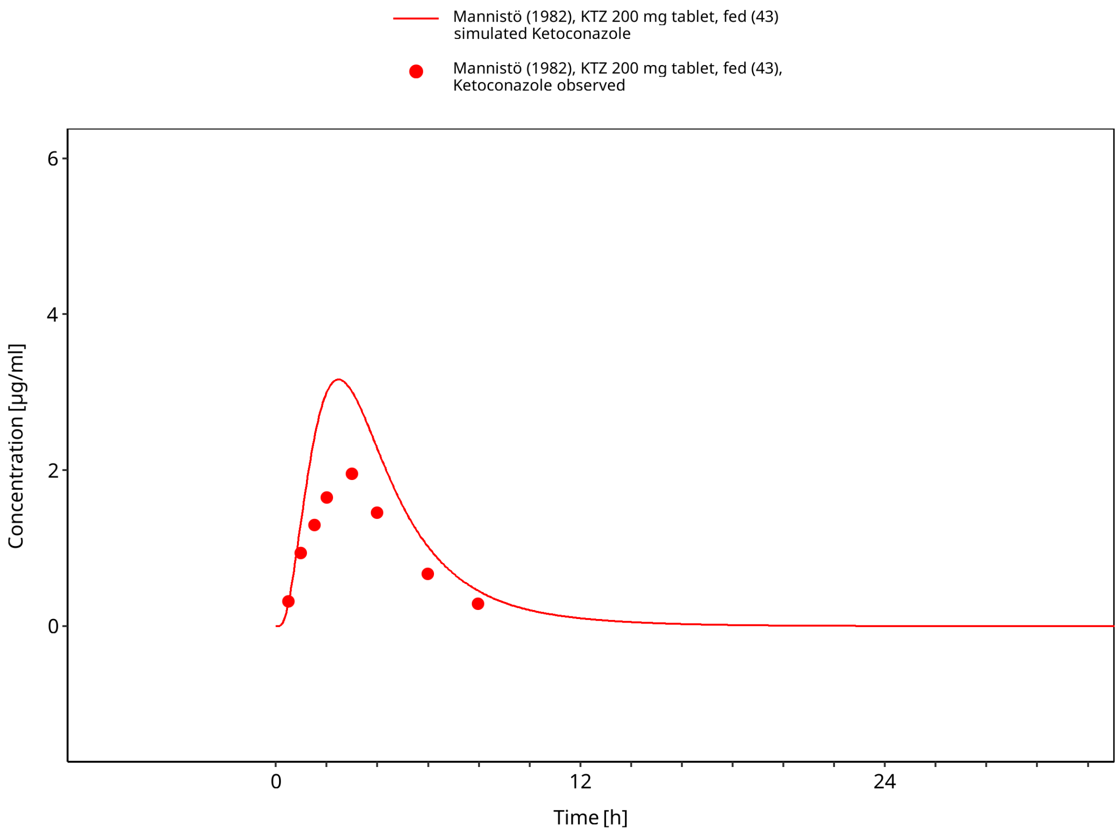

Figure 3-47: Time Profile Analysis

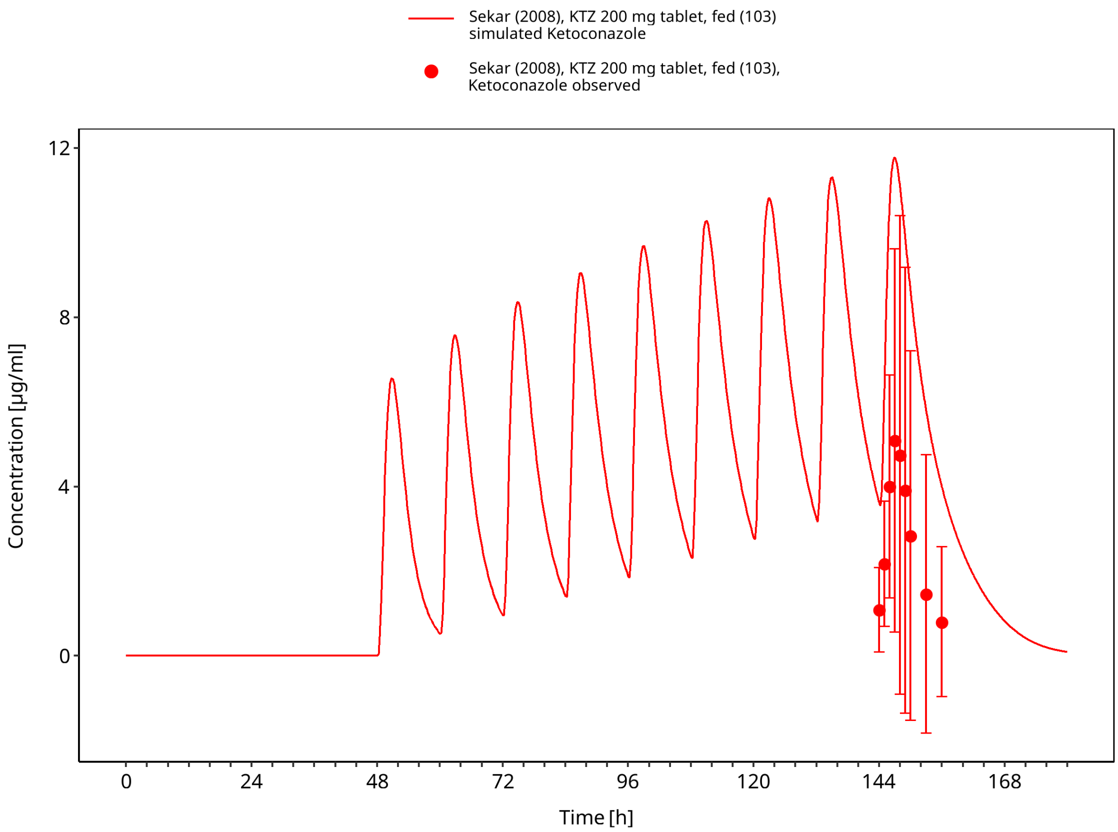

Figure 3-48: Time Profile Analysis

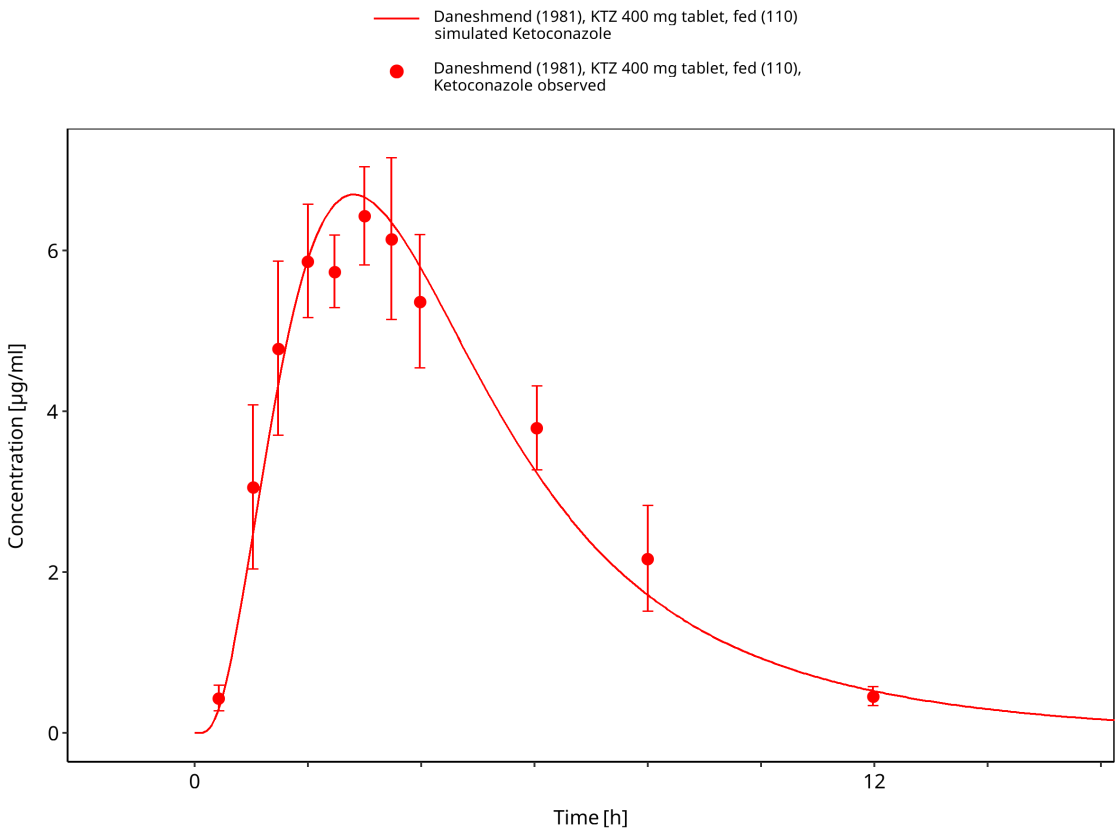

Figure 3-49: Time Profile Analysis

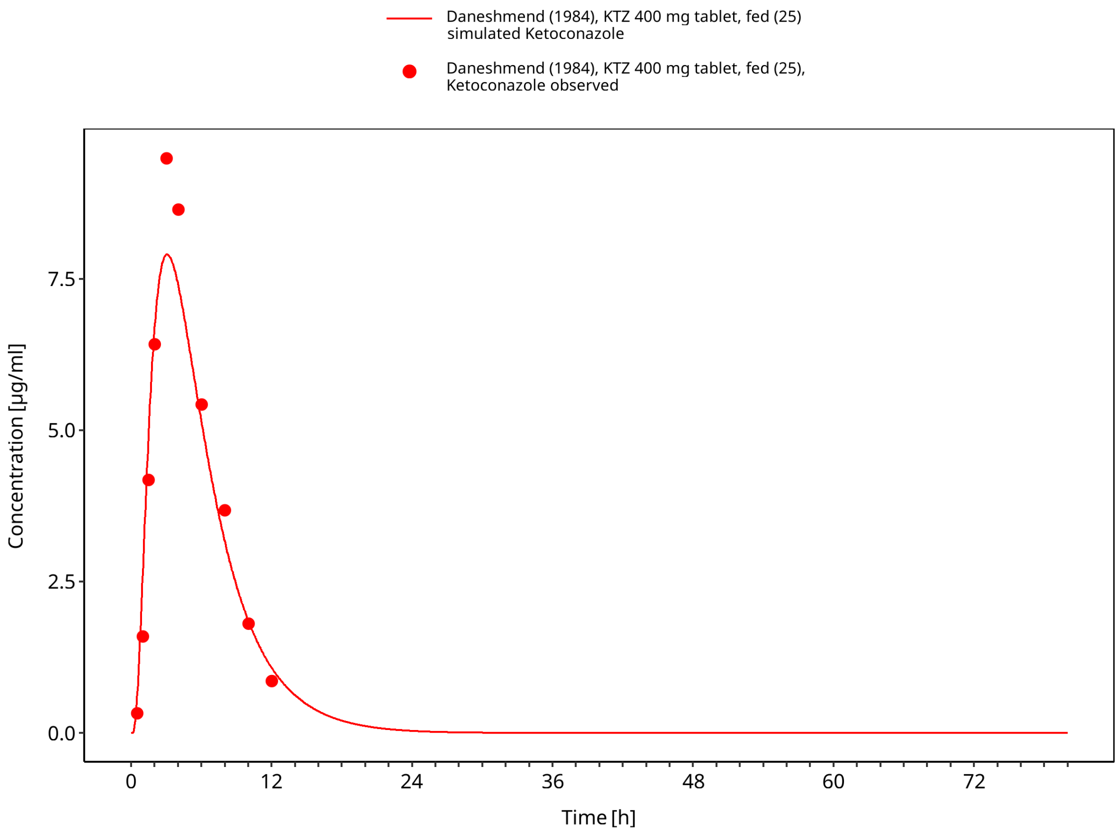

Figure 3-50: Time Profile Analysis

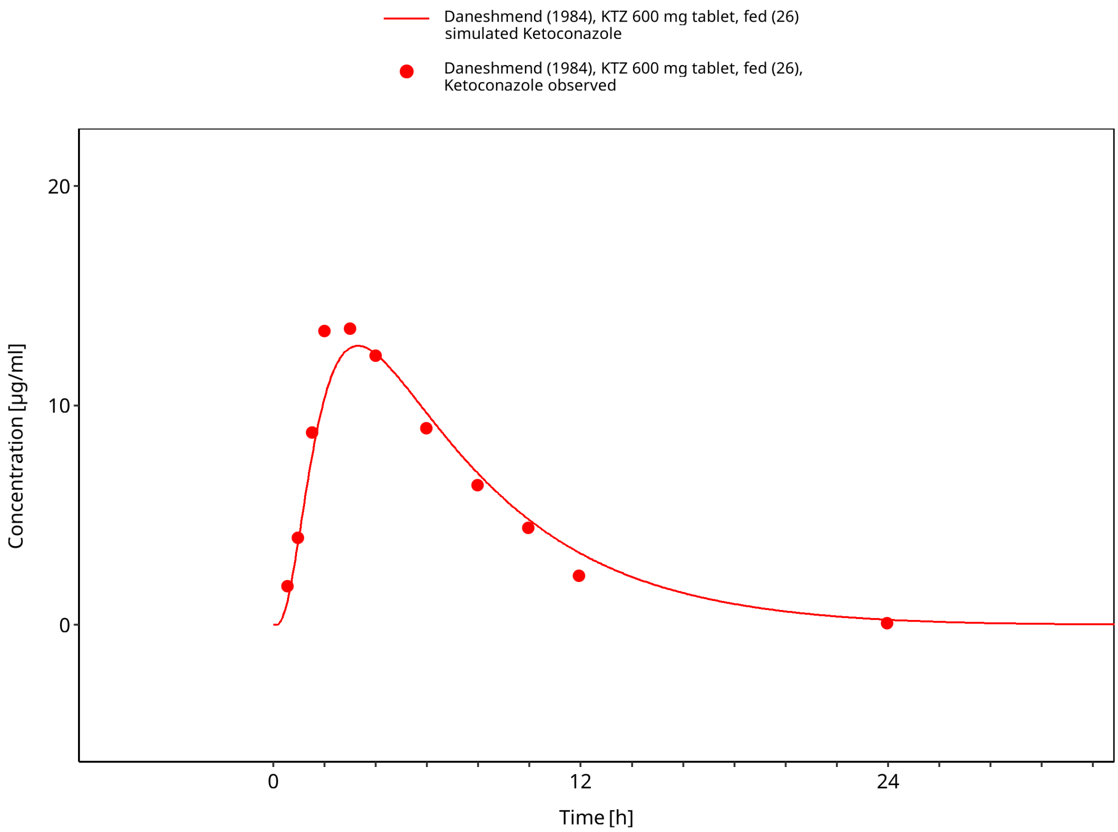

Figure 3-51: Time Profile Analysis

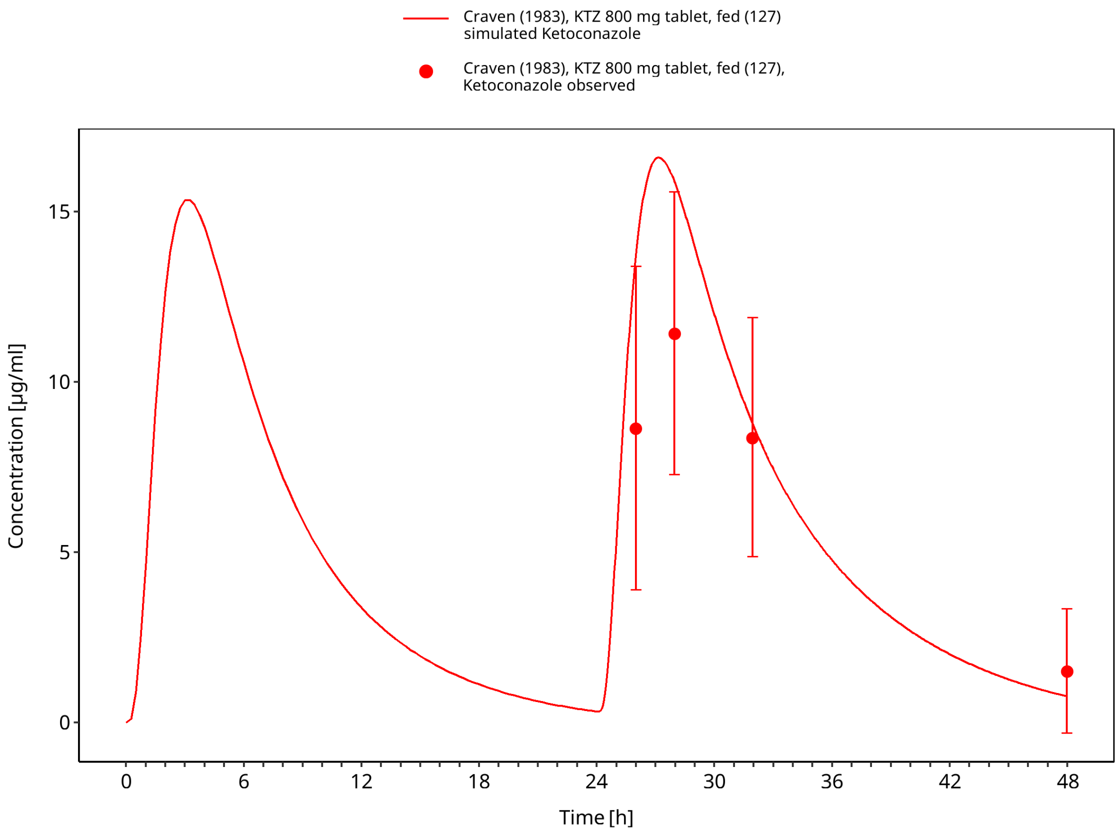

Figure 3-52: Time Profile Analysis

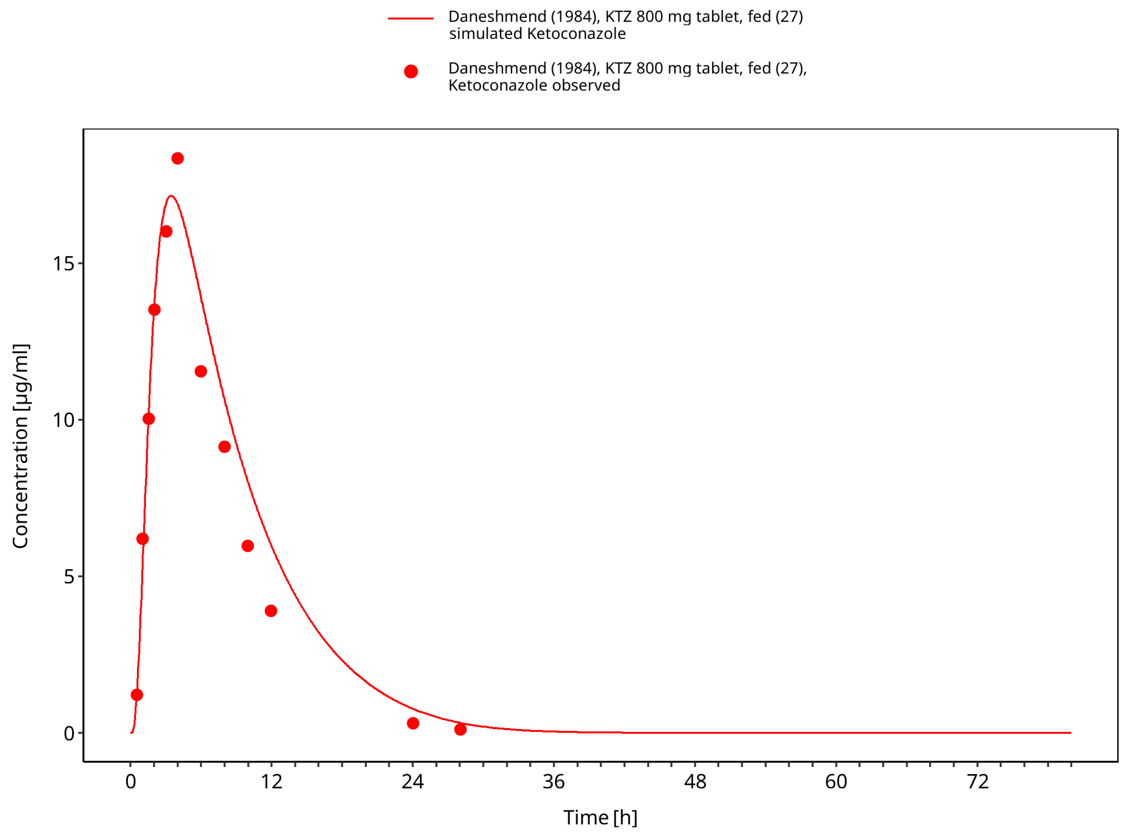

Figure 3-53: Time Profile Analysis

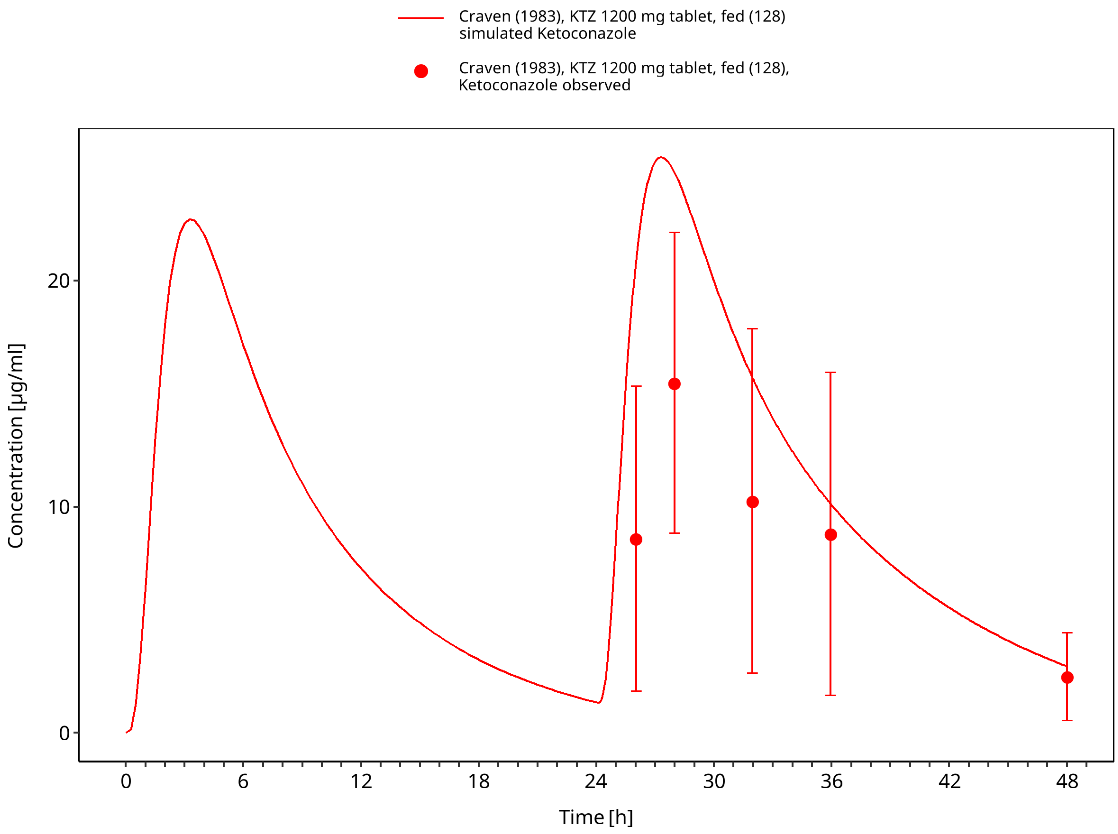

Figure 3-54: Time Profile Analysis

3.3.4 Multiple-dose studies

This section contains multiple-dose oral ketoconazole studies used to evaluate accumulation and time-dependent exposure under repeated dosing.

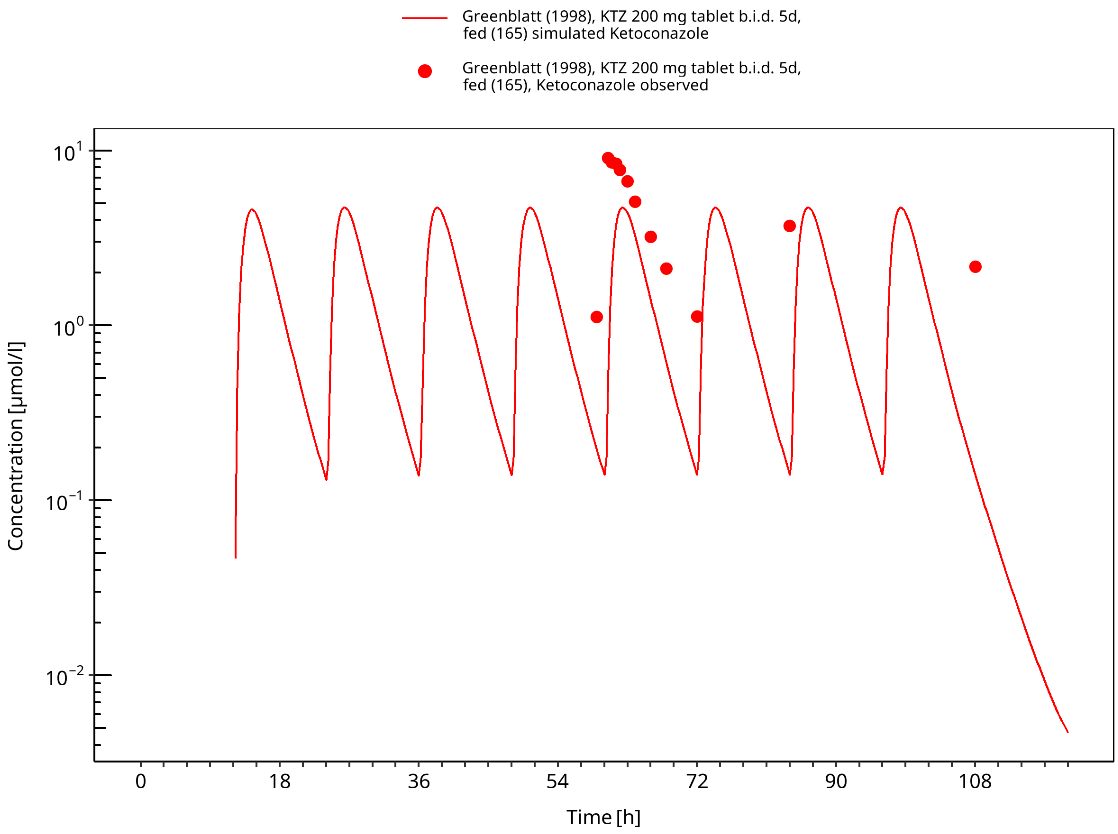

Figure 3-55: Time Profile Analysis

4 Conclusion

The ketoconazole parent-metabolite PBPK model provides a mechanistic description of oral ketoconazole pharmacokinetics across formulation, food-effect and multiple-dose scenarios. The model structure supports DFI and DDI perpetrator applications by representing ketoconazole and its metabolites as contributors to CYP3A4 and P-gp inhibition.

5 References

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1. The snapshot stores the ketoconazole solubility relationship as a table function. The scalar displayed in the generated parameter table is an internal representation of that function and should not be interpreted as the solubility at a specific pH.↩